A nonabelian Fourier transform for unipotent representations of exceptional p-adic groups

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Abstract

In this paper, we prove that the elliptic dual nonabelian Fourier transform defined by Aubert–Ciubotaru–Romano in the elliptic space of unipotent representations of a split, simple, simply connected p-adic group G of type F_4 or E_6 is compatible with Lusztig's non-abelian Fourier transform and restrictions to reductive quotients of maximal open compact subgroups. Our work naturally builds on previous work, where this conjectural property of the dual elliptic nonabelian Fourier transform was stated and proven for $G = \mathrm{SL}_n$, PGL_n, SO_{2n+1}, Sp_4 or G_2.

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1 Introduction

The overarching aim of the Local Langlands Correspondence (LLC) is to classify the smooth admissible complex representations of reductive p-adic groups in terms of geometric and number theoretic data. More concretely, if G is a reductive p-adic group over some non-archimedean field F, the Langlands philosophy predicts that the irreducible (smooth admissible) complex representations $\mathrm{Irr}(G)$ should be controlled by the Weil group $\mathcal{W}_F$ of the field F, together with the geometry of the complex dual group $G^\vee$ of G. In [Lus95a], Lusztig defined the subset of irreducible unipotent representations $\mathrm{Irr}_{\mathrm{un}}(G)$. This is the smallest subset of $\mathrm{Irr}(G)$ containing all irreducible representations containing an Iwahori-fixed vector (also denoted Iwahori-spherical) and closed under the partition of $\mathrm{Irr}(G)$ into L-packets. More concretely, the Local Langlands Correspondence predicts the existence of a bijection,

$$\begin{aligned} \mathrm{LLC}_{\mathrm{un}} : \{(x,\phi) \mid x \in G^\vee, \phi \in \widehat{A_x}\} &\longleftrightarrow \mathrm{Irr}_{\mathrm{un}}(G), \\ (x,\phi) &\longmapsto \pi(x,\phi), \end{aligned} \tag{1}$$

where A_x is the component group of the centralizer $Z_{G^\vee}(x)$, and the left hand side can be seen to be in bijection with unramified Langlands parameters (trivial on the inertia subgroup). In [KL87], Kazhdan–Lusztig had already observed that some of the labels $\{(x,\phi) \mid x \in G^\vee, \phi \in \widehat{A_x}\}$ naturally parametrized Iwahori-spherical representations. The Langlands parametrization for the set of unipotent representations has come after many years of development. Lusztig's work [Lus95a], [Lus02] settled it for adjoint p-adic groups, Reeder [Ree02] for Iwahori-spherical representations of split groups of arbitrary isogeny, and Solleveld for all reductive p-adic groups [Sol23a],[Sol23b]. In addition, Lusztig characterized unipotent representations in terms of the structure of G and representation theory of reductive quotients of its parahoric subgroups, which we revisit in Sections 2 and 3. Deligne–Lusztig theory provides a powerful tool to study representations of finite groups of Lie type and, a posteriori, of p-adic groups.

While many questions in the unipotent LLC have now been resolved in full generality, some of them remain open. One such question relates to the stability of L-packets. Moeglin and Waldspurger tackled this problem for $G = \mathrm{SO}_{2n+1}(F)$ in [MW03] by considering the restriction of unipotent representations to maximal open compact subgroups. These subgroups provide a passage from unipotent representations of the p-adic group to unipotent representations of certain finite reductive groups. Work of Lusztig, Deligne and others in the late 20th century revealed remarkable structure on unipotent characters of these finite reductive groups, which Moeglin and Waldspurger exploited. Lusztig defined an involution on the space of virtual unipotent characters known as *Lusztig's non-abelian transform* taking the irreducible characters to the so-called *almost characters*. These characters can be obtained

independently via the theory of character sheaves and, as a result, have nice geometrical and stability properties.

The hope is that, if we can lift the Fourier transform to the setting of p-adic groups with analogous properties, one could use the resulting virtual characters to prove stability properties of L-packets. Moreover, it is believed by Lusztig that an analogous theory of character sheaves for p-adic groups should exist. Work of Ciubotaru [Ciu20] and Aubert–Ciubotaru–Romano [ACR25] has already considered this problem and formulated precise conjectures. We consider the same setting and similar notation to [ACR25], which we now describe in the simplified setting when G is of simply connected type.

Let $\max(G)$ denote the set of conjugacy classes of maximal open compact subgroups of G, which coincides with the set of conjugacy classes of maximal parahoric subgroups since G is simply connected. For each $K \in \max(G)$, we define the *parahoric restriction map* Does it make much sense to avoid this in the introduction? Feels like I need to introduce U_K, $\overline{K}$ and the representation spaces anyway.

$$\begin{aligned} \mathrm{res}^K_{\mathrm{un}} : R_{\mathrm{un}}(G) &\longrightarrow R_{\mathrm{un}}(\overline{K}) \\ V &\longmapsto V^{U_K}, \end{aligned}$$

where U_K is the pro-p unipotent radical K, $\overline{K} := K/U_K$ is a connected finite group of Lie type and $R_{\mathrm{un}}(G)$ and $R_{\mathrm{un}}(\overline{K})$ are the complexification of the Grothendieck groups spanned by the unipotent representations of G and $\overline{K}$, respectively. Secondly, Lusztig's nonabelian Fourier transforms on $R_{\mathrm{un}}(\overline{K})$ for each $K \in \max(G)$ gives an involution

$$\mathrm{FT}^{\mathrm{cpt}}_{\mathrm{un}} : \bigoplus_{K \in \max(G)} R_{\mathrm{un}}(\overline{K}) \longrightarrow \bigoplus_{K \in \max(G)} R_{\mathrm{un}}(\overline{K}).$$

The remaining ingredient to formally state Aubert–Ciubotaru–Romano's conjecture [ACR25, Conjecture 1.2] is the construction of the *dual nonabelian Fourier transform* $\mathrm{FT}^\vee : R_{\mathrm{un}}(G) \longrightarrow R_{\mathrm{un}}(G)$ compatible with the unipotent restriction maps $\mathrm{res}^K_{\mathrm{un}}$ and Lusztig's nonabelian Fourier transform $\mathrm{FT}^{\overline{K}}$. The existence of such a map is conjectured, but so far no explicit general construction has been achieved on the infinite-dimensional space $R_{\mathrm{un}}(G)$. Instead, Ciubotaru–Opdam [CO17] realized that, for applications, it was often sufficient to study the quotient space $\overline{R}_{\mathrm{un}}(G)$ of $R_{\mathrm{un}}(G)$ by the span of all properly parabolically induced representations, since parabolic induction is generally well understood. The upshot of this simplification is double: firstly, the quotient space $\overline{R}_{\mathrm{un}}(G)$ is finite dimensional, in contrast to the infinite-dimensional space $R_{\mathrm{un}}(G)$. Secondly, $\overline{R}_{\mathrm{un}}(G)$ is canonically isomorphic (in fact, isometric with respect to some canonical inner products) to the elliptic unipotent representation space $R_{\mathrm{un,ell}}(G)$, a certain subspace of $R_{\mathrm{un}}(G)$. Moreover, the Local Langlands Correspondence provides an explicit description for a natural orthogonal basis of $R_{\mathrm{un,ell}}(G)$. To describe this basis, let $u \in G^\vee$ be a unipotent element and let Γ_u be the reductive part of the centralizer $Z_{G^\vee}(u)$. We then consider the

set of elliptic pairs

$$\mathcal{Y}(\Gamma_u)_{\mathrm{ell}} = \{(s,h) \in \Gamma_u \mid sh = hs \text{ and } Z_{\Gamma_u}(s,h) \text{ finite}\}$$

and, for each such elliptic pair (s,h), we define the virtual characters

$$\Pi(u,s,h) := \sum_{\phi \in A_{us}} \phi(h)\pi(us,\phi).$$

The Local Langlands Correspondence induces an isomorphism

$$\mathrm{LLC}^{\mathrm{ell}}_{\mathrm{un}} : \bigoplus_{u \in G^\vee} \mathbb{C}[\mathcal{Y}(\Gamma_u)]^{\Gamma_u} \longrightarrow R_{\mathrm{un,ell}}(G), \qquad (u,s) \longmapsto \Pi(u,s,h),$$

where the sum in the left hand side is taken over conjugacy classes of unipotent elements in $G^\vee$, so in particular $R_{\mathrm{un,ell}}(G)$ is the span of $\{\Pi(u,s,h) \mid u \in G^\vee \text{ unipotent}, (s,h) \in \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}\}$, where the parameters are taken up to $G^\vee$-conjugation.

The space of elliptic pairs has an obvious involution given by the flip $(s,h) \mapsto (h,s)$, and this defines an involution

$$\mathrm{FT}^\vee_{\mathrm{ell}} : R_{\mathrm{un,ell}}(G) \longrightarrow R_{\mathrm{un,ell}}(G),$$

the dual elliptic nonabelian Fourier transform. We are finally ready to state the main conjecture of this paper, in the simplified setting of split, simply connected p-adic groups.

Conjecture 1.1. [ACR25, Conjecture 1.2] Let G be a split semisimple simply connected p-adic group. Then the diagram

$$\begin{array}{ccc} R_{\mathrm{un,ell}}(G) & \xrightarrow{\mathrm{FT}^\vee_{\mathrm{ell}}} & R_{\mathrm{un,ell}}(G) \\ \downarrow{\scriptstyle \mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}} & & \downarrow{\scriptstyle \mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}} \\ \bigoplus_{K \in \max(G)} R_{\mathrm{un}}(\overline{K}) & \xrightarrow{\mathrm{FT}^{\mathrm{cpt}}_{\mathrm{un}}} & \bigoplus_{K \in \max(G)} R_{\mathrm{un}}(\overline{K}), \end{array}$$

commutes, up to certain roots of unity.

In [ACR25], Aubert–Ciubotaru–Romano stated a more general conjecture including all isogeny classes, but the technical details are unnecessary for our purposes. Moreover, they showed that this conjecture (or the more general version) holds for $G = \mathrm{SL}_n(F)$, $\mathrm{PGL}_n(F)$ or $\mathrm{Sp}_4(F)$. In previous work, Ciubotaru and Opdam [Ciu20] showed it holds for $G_2(F)$, and Waldspurger [Wal18] showed it for $G = \mathrm{SO}_{2n+1}(F)$. In this document we settle the above conjecture for two further exceptional cases.

Theorem 1.2. *Let G be a simple, split, simply connected p-adic group of type F_4 or E_6. Then Conjecture 1.1 holds for G.*

Structure of the document

The document is organized as follows. We devote Section 2 to finite groups of Lie type and Deligne–Lusztig theory, with a focus on the construction of Lusztig's nonabelian Fourier transform. In Section 3 we discuss representation theoretic aspects of p-adic groups and their relation to representations of finite groups of Lie type. We recall the definition of unipotent representations and construct the dual nonabelian Fourier transform. Section 4 is the most technical; there we discuss in detail how to explicitly compute the restriction functors $\mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}$, a method developed by Reeder in [Ree00] for adjoint groups and which we adapt for simply connected groups. Sections 5 and 6 contain geometric information about the complex reductive groups $G^\vee = F_4(\mathbb{C})$ or $E_6(\mathbb{C})_{\mathrm{sc}}$ necessary to explicitly describe the virtual representations $\Pi(u, s, h)$ spanning the elliptic unipotent representation space $R_{\mathrm{un},\ell}(G)$.

In Section 7, we display all tables containing the parahoric restrictions of unipotent representations with respect to maximal parahoric subgroups. From these tables, the proof of Conjecture 1.1 is a simple verification.

2 Unipotent representations for finite groups of Lie type

In this first section, we recall the necessary structural and representation theoretic properties of finite groups of Lie type leading to Lusztig's construction of the nonabelian Fourier transform described in the introduction. To this aim, suppose that $\mathbb{G}$ is a connected reductive group over $\overline{\mathbb{F}}_p$ with Frobenius map $\mathrm{Fr} : \mathbb{G} \to \mathbb{G}$. The group $\mathbb{G}^{\mathrm{Fr}}$ is a finite group of Lie type and we are interested in studying the set of irreducible unipotent representations of $\mathbb{G}^{\mathrm{Fr}}$ denoted by $\mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})$. We say that $\mathbb{G}^{\mathrm{Fr}}$ is *split* if the induced Frobenius action on the set of simple roots of $\mathbb{G}$ is trivial. We shall assume throughout that $\mathbb{G}^{\mathrm{Fr}}$ is a split group. We recall that Deligne–Lusztig induction associates, to each pair $(\mathbb{T}, \theta)$ where $\mathbb{T}$ is a Fr-stable maximal tori of $\mathbb{G}$ and θ is a character of $\mathbb{T}^{\mathrm{Fr}}$, a virtual character $R_{\mathbb{T},\theta}$. Then we define

$$\mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}}) = \{\rho \in \mathrm{Irr}(\mathbb{G}^{\mathrm{Fr}}) \mid \langle \rho, R_{\mathbb{T},\mathbf{1}} \rangle \neq 0 \text{ for some Fr-stable torus } \mathbb{T} \text{ of } \mathbb{G}\}.$$

If the Fr-maximal torus $\mathbb{T}$ is maximally split and therefore contained in some Fr-stable Borel subgroup $\mathbb{B}$, then $R_{T_e,\mathbf{1}} = \mathrm{Ind}_{\mathbb{B}^{\mathrm{Fr}}}^{\mathbb{G}^{\mathrm{Fr}}} \mathbf{1}$, so the set of unipotent characters contain the principal series representations of $\mathbb{G}^{\mathrm{Fr}}$. By results of Lusztig, the classification of $\mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})$ is reduced to the case when $\mathbb{G}$ is adjoint simple.

Let $W = N_{\mathbb{G}}(\mathbb{T})/\mathbb{T}$ be the Weyl group associated to $\mathbb{G}$. When the group $\mathbb{G}^{\mathrm{Fr}}$ is split, irreducible representations ϕ of W are known to to parametrize principal series representations χ_ϕ of $\mathbb{G}^{\mathrm{Fr}}$, and we shall make use of this bijection throughout. maybe mention here that other representation series are also in bijection with weyl groups of smaller types?. In addition, for each $\phi \in \widehat{W}$, define the *almost character*

$$R_\phi := \frac{1}{|W|} \sum_{w \in W} \phi(w) R_{\mathbb{T}_w,1},$$

arising as traces of certain character sheaves on the group $\mathbb{G}$. One can then partition $\mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})$ into families by defining two representations ρ_1, ρ_2 to be related if

$$\langle \rho_1, R_\phi \rangle \neq 0 \quad \text{and} \quad \langle \rho_2, R_\phi \rangle \neq 0 \quad \text{for some } \phi \in \widehat{W},$$

and then extending by transitivity. To each family $\mathcal{U} \subset \mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})$, Lusztig associated a finite group $\Gamma_{\mathcal{U}}$, scalars $\Delta(x,\sigma) \in \{\pm 1\}, (x,\sigma) \in \mathcal{M}(\Gamma_{\mathcal{U}})$ defined in [Lus84] and a bijection

$$\begin{aligned} \mathcal{U} &\longleftrightarrow \mathcal{M}(\Gamma_{\mathcal{U}}) := \{(x,\sigma) \mid x \in \Gamma_{\mathcal{U}}, \sigma \in \mathrm{Irr}(Z_{\Gamma_{\mathcal{U}}}(x))\}/\Gamma_{\mathcal{U}} \\ \rho_{(x,\sigma)} &\longleftrightarrow (x,\sigma). \end{aligned}$$

The indexing set $\mathcal{M}(\Gamma_{\mathcal{U}})$ carries a natural pairing

$$\{(x,\sigma),(y,\tau)\} = \frac{1}{|Z_{\Gamma_{\mathcal{U}}}(x)||Z_{\Gamma_{\mathcal{U}}}(y)|} \sum_{\substack{g \in \Gamma_{\mathcal{U}} \\ xgyg^{-1} = gygx^{-1}}} \sigma(gyg^{-1})\tau(g^{-1}x^{-1}g),$$

and the above bijection satisfies the property that

$$\langle \rho_{(x,\rho)}, R_\phi \rangle_{\mathbb{G}^{\mathrm{Fr}}} = \begin{cases} \Delta(x,\sigma)\,\{(x,\rho),(y,\tau)\} & \text{if } \chi_\phi = \rho_{(y,\tau)} \in \mathcal{U}, \\ 0 & \text{if } \chi_\phi \notin \mathcal{U}. \end{cases} \tag{2}$$

We recall that if W is an irreducible Weyl group, then $\Delta(x,\rho) = 1$ except for the families containing $\phi_{512,11}$ in E_7, $\phi_{4096,11}$ in E_8 or $\phi_{4096,26}$ in E_8. For any family $\mathcal{U}$, $\rho_{(e,1)} = \chi_\phi$ is a principal series representation corresponding to some representation ϕ of W. Such representations of W are called *special*, and they are the only ones satisfying the property that $\langle \chi, R_\phi \rangle \geq 0$ for all unipotent characters χ (see [Car85, §12.3]). We define the *unipotent almost characters* of $\mathbb{G}^{\mathrm{Fr}}$ to be the set of orthonormal class functions

$$R_{(x,\sigma)} := \sum_{(y,\tau)} \Delta(y,\tau)\{(y,\tau),(x,\sigma)\}\rho_{(y,\tau)}, \quad (x,\sigma) \in \mathcal{M}(\Gamma_{\mathcal{U}}).$$

In particular, if $\chi_\phi = \rho_{(x,\rho)}$, then $R_\phi = R_{(x,\rho)}$. Lusztig also defined the map

$$\begin{aligned} \mathrm{FT}_{\Gamma_{\mathcal{U}}} : \mathbb{C}[\mathcal{U}] &\longrightarrow \mathbb{C}[\mathcal{U}] \\ \rho_{(x,\sigma)} &\longmapsto \sum_{(y,\tau)\in\mathcal{M}(\Gamma_{\mathcal{U}})} \{(x,\sigma),(y,\tau)\}\rho_{(y,\tau)} \end{aligned}$$

The *unipotent nonabelian Fourier transform* of $\mathbb{G}^{\mathrm{Fr}}$ on the space $R_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}}) = \oplus_{\mathcal{U}\subseteq \mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})}\mathbb{C}[\mathcal{U}]$ is the map

$$\mathrm{FT}^{\mathbb{G}^{\mathrm{Fr}}} := \bigoplus_{\mathcal{U}\subset \mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})} \mathrm{FT}_{\Gamma_{\mathcal{U}}},$$

and it satisfies many interesting properties. Most importantly, it is a self-Hermitian involution with respect to the standard inner product of characters and, in addition, $\mathrm{FT}^{\mathbb{G}^{\mathrm{Fr}}}(\chi_\phi) = R_\phi$. More generally, it takes unipotent representations to the almost characters arising as traces of character sheaves on $\mathbb{G}$. Let us discuss some important examples that will be used in our tables in Section 7.

Example 2.1. When $\mathbb{G} = \mathrm{PGL}_n(\overline{\mathbb{F}_p})$ and Fr is the standard q-th power Frobenius, then $\mathbb{G}^{\mathrm{Fr}} = \mathrm{PGL}_n(\mathbb{F}_q)$ and $W \cong S_n$. In this case, one can show that $R_\phi = \chi_\phi$ for all $\phi \in \widehat{S_n}$ so, in particular, all families have a unique element and unipotent representations coincide with the principal series. Thus, unipotent representations of $\mathbb{G}^{\mathrm{Fr}}$ are naturally by partitions of n, and we shall use this notation in the tables from Section 7.

Example 2.2. Suppose $\mathbb{G}^{\mathrm{Fr}} = \mathrm{SO}_{2n+1}(\mathbb{F}_q)$ or $\mathbb{G}^{\mathrm{Fr}} = \mathrm{PGSp}_{2n}(\mathbb{F}_q)$ (split, types B_n or C_n), and with Weyl group $W \cong S_n \rtimes C_2^n$. Its representations are parametrized by double partitions of n, and we shall use this notation to label principal series representations. In turn, double partitions of n are in bijection with symbols

$$\begin{pmatrix} \lambda_1, & \lambda_2, & \lambda_3, \ldots, & \lambda_{k+s} \\ & \mu_1, & \mu_2, & \ldots, \mu_k \end{pmatrix}$$

with $0 \leq \lambda_1 < \lambda_2 < \cdots < \lambda_{k+s}$, $0 \leq \mu_1 < \mu_2 < \cdots < \mu_k$, with $\lambda_1\mu_1 \neq 0$, rank n (should define rank

or refer to a definition) and defect $s = 1$. To account for the remaining unipotent representations of $\mathbb{G}^{\mathrm{Fr}}$, we allow for symbols with positive odd defect s. Remarkably, Lusztig showed that two unipotent representations lie in the same family if and only if the unordered set $\{\lambda_1, \dots, \lambda_{k+s}, \mu_1, \dots, \mu_k\}$ coincide. See [Car85, §11.4 and §13.8] for a complete overview of the theory.

For example, there are 10 double partitions of $n = 3$ (see Table 12) and the two symbols $\theta_1 = \left(\begin{smallmatrix} 0 & 1 & 2 & 3 \\ & 1 & & \end{smallmatrix}\right)$ and $\theta_2 = \left(\begin{smallmatrix} 0 & 1 & 3 \\ & - & \end{smallmatrix}\right)$ are the only two rank-3 symbols with higher defect (see Table 13). The 12 unipotent representations are partitioned in 4 trivial families and 2 families of size 4 containing the symbols

$$\left\{[1,11] = \left(\begin{smallmatrix} 0 & 1 & 3 \\ 1 & 2 & \end{smallmatrix}\right),\ [\cdot,12] = \left(\begin{smallmatrix} 0 & 1 & 2 \\ 1 & 3 & \end{smallmatrix}\right),\ [1^3,\cdot] = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 0 & 1 & \end{smallmatrix}\right),\ \theta_1 = \left(\begin{smallmatrix} 0 & 1 & 2 & 3 \\ & 1 & & \end{smallmatrix}\right)\right\} \quad \text{and}$$
$$\left\{[2,1] = \left(\begin{smallmatrix} 0 & 3 \\ 1 & \end{smallmatrix}\right),\ [\cdot,3] = \left(\begin{smallmatrix} 0 & 1 \\ 3 & \end{smallmatrix}\right),\ [12,\cdot] = \left(\begin{smallmatrix} 1 & 3 \\ 0 & \end{smallmatrix}\right),\ \theta_2 = \left(\begin{smallmatrix} 0 & 1 & 3 \\ & - & \end{smallmatrix}\right)\right\}$$

When $n = 4$, there are 20 double partitions (see Table 5) and 5 rank-4 symbols with higher defect, namely $\theta_1 = \left(\begin{smallmatrix} 0 & 1 & 2 & 3 & 4 \\ & 1 & 2 & & \end{smallmatrix}\right)$, $\theta_2 = \left(\begin{smallmatrix} 0 & 1 & 2 & 3 \\ & & 2 & & \end{smallmatrix}\right)$, $\theta_3 = \left(\begin{smallmatrix} 0 & 1 & 2 & 4 \\ & 1 & & \end{smallmatrix}\right)$, $\theta_4 = \left(\begin{smallmatrix} 0 & 2 & 3 \\ & - & \end{smallmatrix}\right)$ and $\theta_5 = \left(\begin{smallmatrix} 0 & 1 & 4 \\ & - & \end{smallmatrix}\right)$ (see Table 6). These unipotent representations all lie in a family of size 4 whose associated group is $\Gamma \cong C_2$. As an example, the Fourier transform for these families is

$$\begin{pmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 & -1/2 \\ 1/2 & -1/2 & 1/2 & -1/2 \\ 1/2 & -1/2 & -1/2 & 1/2 \end{pmatrix}.$$

Example 2.3. Consider the split finite group of Lie type $\mathbb{G}^{\mathrm{Fr}} = F_4(\mathbb{F}_q)$. The Weyl group W is a Coxeter group of order 1152 and 25 irreducible representations labelled as $\phi_{d,k}$, where d is the dimension and k is its b-invariant. The outer automorphism of W swapping long and short roots acts on $\widehat{W}$ while preserving these invariants, so for some pairs d, k one has two inequivalent representations labelled by $\phi'_{d,k}$ and $\phi''_{d,k}$. In addition to the principal series representations, $\mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})$ contains 7 cuspidal representations labelled by $F_4^I[1], F_4^{II}[1], F_4[-1], F_4[\theta], F_4[\theta^2], F_4[i]$ and $F_4[-i]$. Finally, it contains 5 more unipotent representations arising as irreducible components of the parabolic induction of the unique unipotent cuspidal representation of a Levi of type B_2, labelled as $B_{2,\mathbf{1}}$, $B_{2,\varepsilon}$, $B_{2,\varepsilon'}$, $B_{2,\varepsilon''}$ and $B_{2,r}$.

The set $\mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})$ is then partitioned in 11 families: there are 8 families consisting of a single representation, 2 families $\{\phi_{4,1}, \phi'_{2,4}, \phi''_{2,4}, B_2, \mathbf{1}\}$, $\{\phi_{4,13}, \phi'_{2,16}, \phi''_{2,16}, B_{2,\varepsilon}\}$ whose associated group is $\Gamma \cong C_2$ and labels $(1,1), (1,\varepsilon), (g_2,1), (g_2,\varepsilon)$, and one large family of size 21 whose associated group is $\Gamma \cong S_4$. See Table 1 for the representations and their labels in $\mathcal{M}(S_4)$.

$\phi_{12,4}$	$\phi''_{9,6}$	$\phi'_{9,6}$	$\phi''_{1,12}$	$\phi'_{1,12}$	$\phi''_{4,7}$	$\phi'_{4,7}$	$\phi_{4,8}$	$\phi'_{6,6}$	$\phi''_{6,6}$	$\phi_{16,5}$	$B_{2,r}$	$B_{2,\varepsilon'}$	$B_{2,\varepsilon''}$	$F_4[\theta]$	$F_4[\theta^2]$	$F_4[i]$	$F_4[-i]$	$F_4^I[1]$	$F_4^{II}[1]$	$F_4[-1]$
$(1,1)$	$(g_2',1)$	$(1,\lambda^1)$	(g_2',ε')	$(1,\lambda^2)$	$(g_4,1)$	(g_2,ε'')	(g_2',ε'')	$(g_3,1)$	$(1,\sigma)$	$(g_2,1)$	(g_2',r)	$(g_4,-1)$	(g_2,ε')	(g_3,θ)	(g_3,θ^2)	(g_4,i)	$(g_4,-i)$	(g_2',ε)	$(1,\lambda^3)$	(g_2,ε)

Table 1: Largest $F_4(\mathbb{F}_q)$-family of unipotent representations and their labels

When $\mathbb{G}^{\mathrm{Fr}}$ is split (as it is the case throughout this document) it is convenient to express Lusztig's nonabelian Fourier transform in term of stable combinations of unipotent representations. This makes the verification of Conjecture 1.1 cleaner and more straightforward. Let $\mathcal{U}$ be a family of unipotent representations and for each $x \in \Gamma_{\mathcal{U}}$, define the virtual combinations of unipotent characters

$$\Pi_{\mathcal{U}}(x,y) = \sum_{\sigma \in \widehat{Z_{\Gamma_{\mathcal{U}}}(x)}} \sigma(y^{-1})\rho_{(x,\sigma)}, \quad \text{where } y \in Z_{\Gamma_{\mathcal{U}}}(x). \tag{3}$$

Dually, when $\Gamma_{\mathcal{U}}$ is abelian, define the virtual combinations

$$\Pi_{\mathcal{U}}(\sigma,\tau) = \sum_{y \in \Gamma_{\mathcal{U}}} \tau(y)\rho_{(y,\sigma)}, \quad \text{if } \sigma,\tau \in \widehat{\Gamma_{\mathcal{U}}}. \tag{4}$$

Lemma 2.4 ([ACR25, Lemma 5.1])**.** *With the notation of* (3) *and* (4)*,*

$$\mathrm{FT}_{\Gamma_{\mathcal{U}}}(\Pi_{\mathcal{U}}(x,y)) = \Delta(x,y)\Pi_{\mathcal{U}}(y,x), \quad \mathrm{FT}_{\Gamma_{\mathcal{U}}}(\Pi_{\mathcal{U}}(\sigma,\tau)) = \Pi_{\mathcal{U}}(\tau,\sigma),$$

the latter when $\Gamma_{\mathcal{U}}$ *is abelian, where* $\Delta(x,y) = 1$ *unless* $\mathcal{U}$ *is one of the families described in* (2)*, in which case* $\Delta(1,g_2) = \Delta(g_2,1) = -1$ *and* 1 *otherwise.*

3 Unipotent representations of p-adic groups

Throughout this section, let G be a connected simple group defined over a finite extension F with residue field $\mathbb{F}_q$. We further assume that G is split with split maximal torus T and of simply connected type. In particular, G is determined by the p-adic field F and the type of the root system Φ of G. Let $\mathrm{Irr}(G)$ be the set of irreducible smooth representations of G and $\mathrm{Irr}_{\mathrm{un}}(G) \subseteq \mathrm{Irr}(G)$ the subset of unipotent representations. Under the Local Langlands correspondence, there is a bijection

$$\mathrm{LLC}_{\mathrm{un}} : \left\{ \begin{array}{c} \text{Unipotent representations} \\ \text{of } G(F) \end{array} \right\}_{/\cong} \longleftrightarrow \left\{ \begin{array}{c} \text{Pairs } (x,\phi): \\ x \in G^\vee,\ \phi \in \widehat{A_x} \end{array} \right\}_{/G^\vee},$$

By the work of Lusztig, unipotent representations of G can also be characterized directly in terms of the topology of G and the representation theory of finite groups of Lie type. For each facet c in the Bruhat–Tits building $\mathcal{B}(G,F)$, consider the parahoric subgroup $K_c := \mathrm{Stab}_G(c)$. Since, by assumption, G is simply connected, K_c is a connected open compact subgroup of G fitting in a short exact sequence

$$1 \longrightarrow U_c \longrightarrow K_c \longrightarrow \overline{K}_c \longrightarrow 1,$$

where U_c is the pro-unipotent radical of K_c and $\overline{K}_c$ is the group of $\mathbb{F}_q$-rational points of a connected reductive group $\overline{\mathbf{K}}_{\mathbf{c}}$. Since $K_{g\cdot c} = gK_cg^{-1}$ for all facets c of $\mathcal{B}(G,F)$ and $g \in G$, conjugacy classes of parahoric subgroups are in bijection with G-orbits of facets in $\mathcal{B}(G)$. Furthermore, we recall that after fixing a set of simple affine roots S_{aff} of G with corresponding fundamental alcove $\mathcal{C}_0$, there is a bijection

$$\{\text{Subsets } J \subsetneq S_{\mathrm{aff}}\} \longleftrightarrow \{G\text{-orbits of facets in } \mathcal{B}(G)\} \longleftrightarrow \{\text{Conj. classes of parahoric subgroups}\}$$
$$J \longmapsto c_J = \{x \in \overline{\mathcal{C}_0} \mid \langle \alpha, x\rangle \in \mathbb{Z} \iff \alpha \in J, \text{ for any } \alpha \in S_{\mathrm{aff}}\} \longmapsto K_{c_J}$$

satisfying that $J_1 \subseteq J_2$ if and only if $c_{J_2} \subset \overline{c_{J_1}}$ if and only if $K_{c_{J_1}} \subseteq K_{c_{J_2}}$. Throughout, we denote by $(K_J, U_J, \overline{K}_J)$ the parahoric subgroup corresponding to the facet c_J. In particular, maximal parahoric subgroups correspond to facets of minimal dimension and maximal subsets $J \subsetneq S_{\mathrm{aff}}$. When G is simply connected, we recall that the set of maximal parahoric subgroups coincides with the set of maximal open compact groups. We define

$$\max(G) := \{K_x \mid \{x\} \subset \mathcal{B}(G) \text{ facet of minimal dimenson}\}$$

to be the set containing each orbit of maximal parahoric subgroups.

Next, let (π, V) a smooth admissible complex representation of G and let $(K, U, \overline{K})$ be a parahoric subgroup. The space V^{U_K} of fixed points under the pro-unipotent radical is naturally a representation

of $\overline{K} = K/U_K$. This induces the *parahoric restriction functor*

$$\mathrm{res}_K : R(G) \longrightarrow R(\overline{K}), \quad V \longmapsto V^{U_K}, \quad \text{for all } V \in \mathrm{Irr}(G), \tag{5}$$

where $R(G)$ and $R(\overline{K})$ are the complexification of the Grothendieck group of $\mathrm{Irr}(G)$ and $\mathrm{Irr}(\overline{K})$, respectively. If (σ, E) is a cuspidal representation of $\overline{K}$, we define

$$\mathrm{Irr}(G, K, E) = \{(\pi, V) \in \mathrm{Irr}(G) \mid \text{ the } \overline{K}\text{-module } V^{U_K} \text{ contains } E\}.$$

Definition 3.1. We say that an irreducible representation (π, V) of G is *unipotent* if there is a parahoric subgroup $(K, U_K, \overline{K})$ such that V^{U_K} contains a cuspidal unipotent representation of $\overline{K}$; that is, if $(\pi, V) \in \mathrm{Irr}(G, K, E)$ for some pair (K, E) where E is unipotent.

Therefore, the set of unipotent representations, up to isomorphism, is

$$\mathrm{Irr}_{\mathrm{un}}(G) = \bigcup_{\substack{J \subsetneq S_{\mathrm{aff}} \\ E \text{ cusp. unip. } \overline{K}_J\text{-rep}}} \mathrm{Irr}(G, K_J, E).$$

Each set $\mathrm{Irr}(G, K, E)$ generates a distinct Bernstein *block* of representations of G. If $\mathcal{I}$ is an Iwahori subgroup, $\mathcal{I}/U_{\mathcal{I}} \cong (\mathbb{F}_q^\times)^r$ and the only unipotent is the trivial character. Thus, Iwahori-spherical representations are all unipotent. Moreover, unipotent representations behave well under the parahoric restriction functors res^K. The first important result is due to Moy and Prasad. A similar result holds for non-unipotent representations, but it is more complicated.

Proposition 3.2. [Ree00, Lemma 5.1] *Assume G is simply connected. Suppose $I \subsetneq S_{\mathrm{aff}}$ and that V^{U_I} contains a cuspidal unipotent representation σ of $\overline{K}_I$. If $J \subsetneq S_{\mathrm{aff}}$ with $V^{U_J} \neq 0$, and J is minimal with respect to this property $I = J$.*

The following is a direct consequence of the above result.

Corollary 3.3. *For any two pairs (K, E) and (K', E') of a parahoric subgroup and a cuspidal unipotent representation of the reductive quotient, $\mathrm{Irr}(G, K, E)$ and $\mathrm{Irr}(G, K', E')$ are either disjoint or equal.*

Proposition 3.2 together with a standard Harish–Chandra style argument yields the following corollary.

Corollary 3.4. *Let (π, V) be a unipotent representation of G and let $(K, U_K, \overline{K})$ be a parahoric subgroup. Then the $\overline{K}$-irreducible components of V^{U_K} are all unipotent.*

Consequently, for any parahoric subgroup K, the parahoric restriction functor res^K naturally restricts to the *unipotent parahoric restriction function*

$$\mathrm{res}^K_{\mathrm{un}} : R_{\mathrm{un}}(G) \longrightarrow R_{\mathrm{un}}(\overline{K}), \quad V \longmapsto V^{U_K},$$

where $R_{\mathrm{un}}(G)$ and $R_{\mathrm{un}}(\overline{K})$ are the complexification of the Grothendieck groups spanned by $\mathrm{Irr}_{\mathrm{un}}(G)$ and $\mathrm{Irr}_{\mathrm{un}}(\overline{K})$, respectively. Finally, we let

$$\mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}} := \bigoplus_{K\in\max(G)} \mathrm{res}^{K}_{\mathrm{un}} : R_{\mathrm{un}}(G) \longrightarrow \bigoplus_{K\in\max(G)} R_{\mathrm{un}}(\overline{K}).$$

3.1 The Langlands classification and the dual nonabelian Fourier transform

The Local Langlands provides a beautiful classification of the unipotent representations of G in terms of the geometry of the dual group $G^\vee$ of G, of adjoint type. This result is widely known as Lusztig's arithmetic-geometric correspondence.

Theorem 3.5 (The arithmetic-geometric correspondence [Lus95a])**.** *There is a natural bijection*

$$\begin{aligned} G^\vee\backslash\{(x,\phi) \mid x\in G^\vee, \phi\in\widehat{A_x}\} &\longleftrightarrow \mathrm{Irr}_{\mathrm{un}}(G)\\ (x,\phi) &\longmapsto (\pi_{(x,\phi)}, V_{(x,\phi)}). \end{aligned}$$

In particular, there is an isomorphism

$$\mathrm{LLC}_{\mathrm{un}} : \bigoplus_{x\in G^\vee} R(A_x) \longrightarrow R_{\mathrm{un}}(G), \tag{6}$$

where the sum on the left hand side is taken over conjugacy classes of $G^\vee$ and $R(A_x)$ is the space of virtual characters of A_x.

In later sections, we display parahoric restriction tables for representations of G parametrized by several elements $x\in G^\vee$. To do this, we use the following standard notation for the Langlands parameters as in [Ree00], which we now explain.

Notation 3.6. Given a representation $\pi(x,\phi)$, let $x=us$ be the Jordan decomposition. Then $u\in Z_{G^\vee}(s)^0$ is a unipotent element of the connected pseudo-Levi subgroup $Z_{G^\vee}(s)^0$, and ϕ is an irreducible representation of the component group A_{su}. The representation $\pi(x,\phi)$ will be labelled by the type of unipotent element u is inside $Z_{G^\vee}(s)^0$, together with the representation ϕ. For example, if $G^\vee=F_4(\mathbb{C})$ and $u\in G^\vee$ is a unipotent element of type $F_4(a_3)$, there is a semisimple element $s\in Z_{G^\vee}(u)$ of order 2 such that $Z_{G^\vee}(s)$ is connected of type $C_3\times A_1$. The unipotent element u is then subregular in the first component and regular in the second, labelled by $C_3(42)\times A_1$, and $\widehat{A_{su}}=C_2\times C_2$. Thus, the four representations of $F_4(F)$ in the L-packet parametrized by $x=su\in F_4(\mathbb{C})$ are then labelled $[C_3(42)A_1,\pm\pm]$.

Given a Langlands parameter (x,ϕ), it is a natural question to ask inside which unipotent block does the representation $(\pi_{(x,\phi)}, V_{(x,\phi)})$ lies inside of. This is the content of the generalized Springer correspondence, which we will not revise in detail here (see [CMBO25, §4.3]). Instead, we will state a

fundamental result giving a geometric characterization for Iwahori-spherical representations in terms of the data (x, ϕ). These representations lie in the *principal block*, generated by $\mathrm{Irr}(G, \mathcal{I}, \mathbf{1})$. For the other representations that are not Iwahori-spherical, we will use Reeder's computations in [Ree00, §10,11]. To state this result, we note that given the data (x, ϕ), one can consider the partial flag variety $\mathcal{B}^x$ containing all Borel subgroups of $G^\vee$. This variety carries a natural action of A_x given by conjugation. In particular, the cohomology complex $H^*(\mathcal{B}^x)$ inherits an action of A_x too.

Theorem 3.7. [KL87, Theorem 7.12] *The representation $(\pi_{(x,\phi)}, V_{(x,\phi)})$ is Iwahori-spherical if and only if $H^*(\mathcal{B}^x)^\phi \neq 0$. In particular, for any $x \in G^\vee$, the representation $(\pi_{(x,\mathbf{1})}, V_{(x,\mathbf{1})})$ is Iwahori-spherical.*

In this section we also define the dual nonabelian Fourier transform for unipotent representations of p-adic groups. Conjecturally, we expect the existence of a map $\mathrm{FT}^\vee$, a dual nonabelian Fourier transform, on the unipotent representation space $R_{\mathrm{un}}(G)$ with analogous properties to Lusztig's nonabelian Fourier transforms $\mathrm{FT}^{\overline{K}}$ and compatible with the unipotent parahoric restriction maps $\mathrm{res}^K_{\mathrm{un}}$. However, such a map has not been constructed in the literature in general on the infinite dimensional space $R_{\mathrm{un}}(G)$. For some applications, it is enough to construct it on the elliptic unipotent representation space $R_{\mathrm{un,ell}}(G)$, a finite dimensional subspace of $R_{\mathrm{un}}(G)$ which we now describe.

Classically, the elliptic unipotent representation space $\overline{R}_{\mathrm{un}}(G)$ is defined as the quotient of $R_{\mathrm{un}}(G)$ by the subspace spanned by all properly parabolically induced representations. Equivalently, $\overline{R}_{\mathrm{un}}(G)$ is obtained by computing the quotient of $R_{\mathrm{un}}(G)$ by the radical of the Euler–Poincare pairing

$$EP_G(\ ,\ ) : R_{\mathrm{un}}(G) \times R_{\mathrm{un}}(G) \longrightarrow \mathbb{C}, \quad (V, V') \longmapsto \sum_{i \geq 0} (-1)^i \mathrm{Ext}^i(V, V'), \text{ where } V, V' \in \mathrm{Irr}_{\mathrm{un}}(G),$$

and extended by linearity, and where $\mathrm{Ext}^i(V, V')$ are calculated in the category of smooth complex G-representations. Thus, the space $R_{\mathrm{un}}(G)$ carries a natural non-degenerate pairing $EP_G(\ ,\ )$. Similarly, for each $x \in G^\vee$ the space of virtual representations $R(A_x)$ carries a natural elliptic pairing $(\ ,\ )^{A_x}_{\mathrm{ell}}$ (see [ACR25, §8.1] for the definition), and $\overline{R}(A_x)$ is the quotient of $R(A_x)$ by the radical of the elliptic pairing. The arithmetic-geometric (6) induces an isometric isomorphism on the elliptic representation spaces

$$\overline{\mathrm{LLC}}_{\mathrm{un}} : \bigoplus_{x \in G^\vee} \overline{R}(A_x) \longrightarrow \overline{R}_{\mathrm{un}}(G), \tag{7}$$

where the sum on the left is again over conjugacy classes of elements in $G^\vee$. The left hand side of (7) can be further decomposed into

$$\bigoplus_{x \in G^\vee} \overline{R}(A_x) = \bigoplus_{\substack{u \in G^\vee \\ \text{unipotent}}} \ \bigoplus_{\substack{s \in Z_{G^\vee}(u) \\ \text{semisimple}}} \overline{R}(A_{su}).$$

Next, fix some unipotent element $u \in G^\vee$ and let Γ_u be the reductive part of the centralizer $Z_{G^\vee}(u)$.

We define the set of elliptic pairs associated to u to be

$$\mathcal{Y}(\Gamma_u)_{\mathrm{ell}} = \{(s,h) \in \Gamma_u \times \Gamma_u \mid s,h \text{ semisimple}, sh = hs, Z_{\Gamma_u}(s,h) \text{ is finite}\}.$$

It is not hard to show that this set is always finite (see [ACR25, Lemma 8.1]). For every $(s,h) \in \mathcal{Y}(\Gamma_u)$, define

$$\Pi(s,h) = \sum_{\phi \in \widehat{A_{\Gamma_u}(s)}} \phi(\overline{h})\phi \in R(A_{\Gamma_u}(s)), \tag{8}$$

where $\overline{h}$ is the image of h in $A_{\Gamma_u}(s)$. Moreover, let $\overline{\Pi}(s,h)$ denote the image in $\overline{R}(A_{\Gamma_u}(s))$ and let $\mathbb{C}[\mathcal{Y}(\Gamma_u)_{\mathrm{ell}}]^{\Gamma_u} \cong \mathbb{C}[\Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}]$ be the space of Γ_u-invariant functions $\mathcal{Y}(\Gamma_u)$. Let $\mathbf{1}_{[(s,h)]}$ be the indicator function of the Γ_u-orbit of (s,h).

Proposition 3.8. [ACR25, Proposition 8.3] *Let $u \in G^\vee$ be unipotent. Then the correspondence $\mathbf{1}_{[(s,h)]} \mapsto \overline{\Pi}(s,h)$ induces an isomorphism*

$$\mathbb{C}[\mathcal{Y}(\Gamma_u)_{\mathrm{ell}}]^{\Gamma_u} \longrightarrow \bigoplus_{\substack{s \in Z_{G^\vee}(u) \\ \text{semisimple}}} \overline{R}(A_{su})$$

Finally, for each $[(s,h)] \in \Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}$, define the virtual combinations

$$\Pi(u,s,h) := \sum_{\phi \in \widehat{A_{\Gamma_u}(s)}} \phi(h)\pi(su,\phi) \in R_{\mathrm{un}}(G), \tag{9}$$

and let $\overline{\Pi}(u,s,h)$ be the corresponding images in $\overline{R}_{\mathrm{un}}(G)$.

Definition 3.9. The elliptic unipotent representation space $R_{\mathrm{un,ell}}(G)$ is the finite-dimensional subspace of $R_{\mathrm{un}}(G)$ spanned by virtual characters $\{\Pi(u,s,h) \mid u \text{ unipotent, and } (s,h) \in \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}\}$.

Thus, the isomorphism (7) together with Proposition 3.8 yields an isometric isomorphism

$$R_{\mathrm{un,ell}}(G) \longrightarrow \overline{R}_{\mathrm{un}}(G), \quad \Pi(u,s,h) \mapsto \overline{\Pi}(u,s,h) \tag{10}$$

The set $\Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}$ admits an obvious involution given by flip : $(s,h) \mapsto (h,s)$, which induces an involution on $R_{\mathrm{un,ell}}(G)$.

Definition 3.10. The *dual elliptic nonabelian Fourier transform* is the involutive linear map $\mathrm{FT}^\vee_{\mathrm{ell}} : R_{\mathrm{un,ell}} \to R_{\mathrm{un,ell}}$ defined by

$$\mathrm{FT}^\vee_{\mathrm{ell}}(\Pi(u,s,h)) = \Pi(u,h,s), \quad u \in G^\vee \text{ unipotent}, (s,h) \in \Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}.$$

We are now ready to state the main conjecture for elliptic unipotent representations.

Conjecture 3.11. [ACR25, Conjecture 1.2] Let G be a split semisimple p-adic group of simply con-

nected type. Then the following diagram commutes up to roots of unity,

$$\begin{array}{ccc} R_{\mathrm{un,ell}}(G) & \xrightarrow{\mathrm{FT}^{\vee}_{\mathrm{ell}}} & R_{\mathrm{un,ell}}(G) \\ \downarrow{\scriptstyle \mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}} & & \downarrow{\scriptstyle \mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}} \\ \bigoplus_{K\in\max(G)} R_{\mathrm{un}}(\overline{K}) & \xrightarrow{\mathrm{FT}^{\mathrm{cpt}}_{\mathrm{un}}} & \bigoplus_{K\in\max(G)} R_{\mathrm{un}}(\overline{K}), \end{array}$$

where $\mathrm{FT}^{\mathrm{cpt}}_{\mathrm{un}} := \bigoplus_{K\in\max(G)} \mathrm{FT}^{\overline{K}}_{\mathrm{un}}$ on each coordinate.

The horizontal maps in the above conjecture are easily computable. They are involutions on finite dimensional representation spaces with canonical basis, and computing the matrices is straightforward. The difficulty of the above conjecture relies on the computation of the unipotent parahoric restriction maps $\mathrm{res}^{K}_{\mathrm{un}}$. This is due to the fact that $R_{\mathrm{un}}(G)$ has a natural basis in terms of the geometry of the dual group $G^{\vee}$, while irreducible characters $R_{\mathrm{un}}(\overline{K})$ are parametrized by families and Lusztig's labels. The connection between these parametrizations is deep, and we devote the next section to discuss this in detail for Iwahori-spherical representations. At the end we also give a brief discussion on the computation of this map for non-Iwahori-spherical representations. We closely follow Reeder's discussion in [Ree00, §8,9].

4 Parahoric restriction of unipotent representations

Let G be a connected split simple group of simply connected type over a p-adic field F. In the previous section we stated the bijection

$$\begin{aligned} \mathrm{LLC}^p_{\mathrm{un}} : G^\vee \backslash \{(x,\phi) \mid x \in G^\vee, \phi \in \widehat{A_x}\} &\longleftrightarrow \mathrm{Irr}_{\mathrm{un}}(G) \\ (x,\phi) &\longmapsto (\pi_{(x,\phi)}, V_{(x,\phi)}), \end{aligned}$$

parametrizing unipotent representations of G in terms of the geometry of its complex dual group $G^\vee$. For a fixed $J \subsetneq S_{\mathrm{aff}}$ and corresponding parahoric subgroup K_J, in this section we discuss how to explicitly compute the parahoric restriction maps

$$\mathrm{res}^{K_J}_{\mathrm{un}} : R_{\mathrm{un}}(G) \longrightarrow R_{\mathrm{un}}(\overline{K}_J), \quad V \longmapsto V^{U_J}$$

from unipotent representations of G to unipotent representations of $\overline{K}_J$. We closely follow the discussion from [Ree00, §4-8], which we recall for convenience. However, Reeder assumes G to be of adjoint type, and not all results translate as cleanly when G is of simply connected type.

Let us fix throughout a unipotent representation $V = V_{(x,\phi)}$ of G, with $x \in G^\vee$ and $\phi \in \widehat{A_x}$. For each irreducible unipotent $\overline{K}_J$-representation χ, we want to calculate the value of $\langle \chi, V^{U_J} \rangle_{K_J}$, where we view the $\overline{K}_J$-representations as K_J-representations trivial on U_J.

In the first part, we state two reduction steps that allow us to reduce the problem of calculating the parahoric restrictions of V to the problem of calculating the restriction of some representation obtained from V over a reflection group with respect to certain reflection subgroups. This part does not require G to be of adjoint type. For example, if V is Iwahori-spherical, we obtain a representation $V^{\mathbf{1}}_{q=1}$ over the affine Weyl group $W_{\mathrm{aff}} = W \ltimes X(T^\vee)$ of G, and the parahoric restriction of V can be determined by studying the representation $V^{\mathbf{1}}_{q=1}|_{\widetilde{W}_J}$ where $\widetilde{W}_J$ is the subgroup of W_{aff} generated by the reflections in J. Springer theory is then the key theoretical tool that allows us to calculate the action of W_{aff} on $V^{\mathbf{1}}_{q=1}$. We devote the second part of the section to discuss this. However, not every unipotent representation of G is Iwahori-spherical, so in the last part we discuss a more general method to calculate the parahoric restriction of non Iwahori-spherical representations with the use of weight diagrams developed in [Ree97]. These have been computed for all square-integrable unipotent representations of adjoint exceptional groups in [Ree00, §9].

4.1 Reduction to representations over reflection groups

Let $V = V_{(x,\phi)}$ be a unipotent representation of the simply connected p-adic group G. By Corollary 3.3, there is a unique proper subset I of S_{aff} with corresponding standard parahoric $K := K_I$, such

that V^{U_K} contains a cuspidal unipotent representation σ of $\overline{K}$. In particular, the vector space

$$V^{\sigma} := \mathrm{Hom}_K(\sigma, V^{U_K}) = \mathrm{Hom}_K(\sigma, V) = \mathrm{Hom}_G(c\text{-}\mathrm{Ind}_K^G \sigma, V) \tag{11}$$

is a non-zero irreducible $\mathcal{H}(G, K, \sigma) = \mathrm{End}_G(c\text{-}\mathrm{Ind}_K^G \sigma)$-module. The passage $V \mapsto V^{\sigma}$ is known to lose no information about the representation V ([BK98],[Lus95a]).

Similarly, if χ is an irreducible unipotent representation of $\overline{K}_J$, there is a unique proper subset I' of S_{aff} contained in J such that χ is a subrepresentation of $\mathrm{Ind}_{K_{I'}}^{K_J} \sigma'$, where σ' is a cuspidal unipotent representation of $\overline{K}_{I'}$. In particular, the vector space

$$\chi^{\sigma'} := \mathrm{Hom}_{K_{I'}}(\sigma', \chi) = \mathrm{Hom}_{K_J}(\mathrm{Ind}_{K_{I'}}^{K_J} \sigma', \chi) \neq 0$$

is naturally a $\mathcal{H}(K_J, K_{I'}, \sigma')$-module, where $\mathcal{H}(K_J, K_{I'}, \sigma') \cong \mathrm{End}_{K_J}(\mathrm{Ind}_{K_{I'}}^{K_J} \sigma')$ is the subalgebra of functions of $\mathcal{H}(G, K_{I'}, \sigma')$ supported on K_J.

Lemma 4.1 (First reduction)**.** *(*[Ree00, §4.2]*) The multiplicity of the simple K_J-module χ in V^{U_J} is given by*

$$\langle \chi, V^{U_J} \rangle_{K_J} = \langle \chi^{\sigma'}, V^{\sigma'} \rangle_{\mathcal{H}(K_J, K_{I'}, \sigma')}.$$

Moreover, $\langle \chi, V^{U_J} \rangle = 0$ unless $I = I'$. In particular, if J does not contain I, then $V^{U_J} = 0$.

Proof. Since $U_J \subseteq U_{I'}$, we have that

$$\begin{aligned} V^{\sigma'} = \mathrm{Hom}_{K_{I'}}(\sigma', V) = \mathrm{Hom}_{K_{I'}}(\sigma', V^{U_J}) &\simeq \mathrm{Hom}_{K_J}(\mathrm{Ind}_{K_{I'}}^{K_J} \sigma', V^{U_J}) \\ &\simeq \bigoplus_{\eta} \langle \eta, V^{U_J} \rangle_{K_J} \mathrm{Hom}_{K_J}(\mathrm{Ind}_{K_{I'}}^{K_J} \sigma', \eta) \\ &\simeq \bigoplus_{\eta} \langle \eta, V^{U_J} \rangle_{K_J} \eta^{\sigma'}, \end{aligned}$$

and therefore

$$\langle \chi^{\sigma'}, V^{\sigma'} \rangle_{\mathcal{H}(K_J, K_{I'}, \sigma')} = \sum_{\eta} \langle \chi^{\sigma'}, \eta^{\sigma'} \rangle_{\mathcal{H}(K_J, K_{I'}, \sigma')} \langle \eta, V^{U_J} \rangle_{K_J} = \langle \chi, V^{U_J} \rangle_{K_J}.$$

To prove the second statement, we note that $V^{\sigma'} \neq 0$ if and only if $(K, \sigma) = (K_{I'}, \sigma')$ by Corollary 3.3. In other words, $I = I'$ and the last assertion follows immediately from the fact that J contains I'. □

Example 4.2. Let us briefly explore the two extreme cases of the above result.

- If V is a Iwahori-spherical representation of G, then $V^{\mathcal{I}} \neq 0$, $I = \emptyset$ and the pair (K, σ) is given by $(\mathcal{I}, \mathbf{1})$. In this case, Lemma 4.1 states that

$$\langle \chi, V^{U_J} \rangle_{K_J} = \langle \chi^{\mathcal{I}}, V^{\mathcal{I}} \rangle_{\mathcal{H}(K_J, \mathcal{I}, \mathbf{1})}.$$

Thus, V has a non-zero parahoric restriction for any $J \subsetneq S_{\mathrm{aff}}$.

- If V is a supercuspidal unipotent representation of G, then I is a maximal proper subset of S_{aff} and $V^{U_J} = 0$ unless $J = I$.

For the second reduction step we consider $R = \mathbb{C}(v, v^{-1})$, where v is an indeterminate and define $\mathcal{H}(G, K, \sigma)_v$ to be the *generic Hecke algebra* defined over R with the same generators and relations as $\mathcal{H}(G, K, \sigma)$, but with q replaced by v^2 in the quadratic relations. By specializing this generic Hecke algebra we recover

$$\mathcal{H}(G, K, \sigma)_{\sqrt{q}} = \mathcal{H}(G, K, \sigma), \qquad \mathcal{H}(G, K, \sigma)_1 = \mathbb{C}[\tilde{W}(K, \sigma)],$$

where $\tilde{W}(K, \sigma)$ is an explicit affine reflection group associated to the pair (K, σ) (see [Lus95a] for an explicit list, and [Car85, §10] for a detailed construction). When $K = \mathcal{I}$ is the Iwahori subgroup and $\sigma = \mathbf{1}$ is the trivial representation, $\mathcal{H}(G, \mathcal{I}, \mathbf{1})_1 = \mathbb{C}[\widetilde{W}]$, where $\widetilde{W}$ is the affine Weyl group. Instead, if K is a maximal parahoric subgroup of G, then $\mathcal{H}(G, K, \sigma) \cong \mathbb{C}$ so $\tilde{W}(K, \sigma)$ is trivial.

Fact: For any simple $\mathcal{H}(G, K, \sigma)$-module E *considered in this document*, there is a $\mathcal{H}(G, K, \sigma)_v$-module E_v such that

$$E \simeq E_v \otimes_R \mathbb{C}, \text{ where } f \in R \text{ acts on } \mathbb{C} \text{ by } f(\sqrt{q}).$$

It then follows that we have a $q = 1$ operation that takes modules over Hecke algebras to modules over the corresponding reflection groups, obtained by setting $v = 1$ in all matrix coefficients of the generic module.

Proposition 4.3 (Second reduction)**.** *(*[Ree00, §6.1]*) Let $J \subsetneq S_{\mathrm{aff}}$ and let $(K, U_K, \overline{K})$ be a parahoric subgroup contained in $(K_J, U_J, \overline{K}_J)$ with cuspidal unipotent $\overline{K}$-representation σ. Then the diagram*

$$\begin{array}{ccc} \mathcal{H}(G, K, \sigma)\text{-}mod & \xrightarrow{q=1} & \tilde{W}(K, \sigma)\text{-}mod \\ \Big\downarrow{\scriptstyle \mathrm{Res}_{\mathcal{H}(K_J, K, \sigma)}} & & \Big\downarrow{\scriptstyle \mathrm{Res}_{\tilde{W}(K, \sigma)_J}} \\ \mathcal{H}(K_J, K, \sigma)\text{-}mod & \xrightarrow{q=1} & \tilde{W}(K, \sigma)_J\text{-}mod \end{array}$$

is commutative, and the bottom arrow is an isometry with respect to the usual inner product of character. That is, for any irreducible K_J-module χ, we have that

$$\langle \chi^\sigma, V^\sigma \rangle_{\mathcal{H}(K_J, K, \sigma)} = \langle \chi^\sigma_{q=1}, V^\sigma_{q=1} \rangle_{\tilde{W}(K, \sigma)_J}. \tag{12}$$

Combining both reduction steps, we obtain the identity

$$\langle \chi, V^{U_J} \rangle_{K_J} = \langle \chi^\sigma, V^\sigma \rangle_{\mathcal{H}(K_J, K, \sigma)} = \langle \chi^\sigma_{q=1}, V^\sigma_{q=1} \rangle_{\tilde{W}(K, \sigma)_J} \tag{13}$$

Thus, to compute the parahoric restriction V^{U_J}, it suffices to study the $\tilde{W}(K, \sigma)_J$-module $V^\sigma_{q=1}$. By

calculating its decomposition into $\tilde{W}(K,\sigma)$-irreducible representation, one can directly obtain the decomposition of V^{U_J} into irreducible $\overline{K}_J$-representations.

At this point, we restrict our attention to the parahoric restriction of Iwahori-spherical representations, for Springer theory provides a way to calculate the $\tilde{W}(I,\mathbf{1})_J$-module $V^{\mathbf{1}}_{q=1}$ explicitly, differing significantly from the non Iwahori-spherical case. As stated above, for Iwahori-spherical representations, the reflection subgroup $\tilde{W}(\mathcal{I},\mathbf{1}) \cong W_{\text{aff}}$ is simply the affine Weyl group of G, and $\tilde{W}(\mathcal{I},\mathbf{1})_J = \widetilde{W}_J$ is the subgroup of $\widetilde{W}$ generated by the simple reflections of the roots in J.

4.2 The Iwahori-spherical case and the Springer correspondence

Given an Iwahori-spherical representation $V = V_{(x,\phi)}$ of G and a proper subset $J \subsetneq S_{\text{aff}}$, in the previous subsection we proved that

$$\langle \chi, V^{U_J} \rangle_{\overline{K}_J} = \langle \chi^{\mathbf{1}}_{q=1}, V^{\mathbf{1}}_{q=1} \rangle_{\widetilde{W}_J}. \tag{14}$$

for any principal series representation χ of the reductive quotient $\overline{K}_J$. To extract describe the structure of the $\widetilde{W}_J$-module $V^{\mathbf{1}}_{q=1}$ in terms of the geometry of the dual group and the data (x,ϕ), Springer theory provides a powerful tool. Let $x = us$ be the Jordan decomposition of x, with $u \in G^\vee$ unipotent and $s \in G^\vee$ semisimple. By conjugating if necessary, we assume that $s \in T^\vee$ for a fixed split maximal torus of $G^\vee$. Let $\mathcal{B}$ and $\mathcal{B}_s$ be the flag varieties of the complex reductive groups $G^\vee$ and $M^\vee := Z_{G^\vee}(s)^0$, respectively, and consider the Springer fibres $\mathcal{B}^x$ and $\mathcal{B}^u_s$ parametrizing the set of Borel subgroups of $G^\vee$ containing x and the set of Borel subgroups of $M^\vee$ containing u. Let W_M be the Weyl group of $M^\vee$, isomorphic to the subgroup of W generated by all reflections of roots trivial on s.

While the varieties $\mathcal{B}_x$ and $\mathcal{B}^u_s$ do not admit a natural action of W and W_M, respectively, Lusztig showed using the theory of perverse sheaves that the singular cohomology spaces $H^*(\mathcal{B}_x)$ and $H^*(\mathcal{B}^u_s)$ do. The W_M structure of $H^*(\mathcal{B}^u_s)$ can be easily read from the Springer correspondence for the reductive group $M^\vee$ via the Green tables, which are well understood. Moreover, the module structures are tightly related by the formula ([Kat83, Proposition 2.5]),

$$H^*(\mathcal{B}^x) \cong \text{Ind}^W_{W_M} H^*(\mathcal{B}^u_s).$$

Kato also showed that this action could naturally be extended to an action of the affine Weyl group $W_{\text{aff}} = W \ltimes X(T^\vee)$, where $X = X(T^\vee)$ is the character lattice of $T^\vee$. The semisimple element $s \in T^\vee$ is naturally a character of X given by evaluation. Thus, the cohomology spaces $H^*(\mathcal{B}^x)$ carry a natural W_{aff}-action given by

$$H^*(\mathcal{B}^x) = \text{Ind}^{W_{\text{aff}}}_{W_{M,\text{aff}}} H^*(\mathcal{B}^u_s) \otimes s,$$

where $W_{M,\text{aff}} = W_M \ltimes X$ (see [Kat83, Proposition 3.2]).

Finally, the component group A_x acts naturally on the varieties $\mathcal{B}^x$ and $\mathcal{B}^u_s$ by conjugation, inducing

a natural action on the cohomology spaces $H^*(\mathcal{B}^x)$ and $H^*(\mathcal{B}_s^u)$. While the A_x-action on $H^*(\mathcal{B}^x)$ commutes with the W_{aff}-action, thus giving an $A_x \times W_{\text{aff}}$-module structure, the A_x-action on $H^*(\mathcal{B}_s^u)$ does not necessarily commute with the $W_{M,\text{aff}}$-action. The precise structure of this action is subtle, and we closely follow the discussion in [Kat83, §6] to explain this.

Given $x = su$ as above, let W_s be the stabilizer of s in W. Then W_M is a normal subgroup of W_s, and the quotient is the component group of the centralizer of s in $G^\vee$:

$$W_s/W_M \cong Z_{G^\vee}(s)/Z_{G^\vee}(s)^0 \cong R = \{w \in W_s \mid w(\Delta_M^+) = \Delta_M^+\}.$$

By choosing Kac coordinates for s, the group R can be seen as a subgroup of the fundamental group Ω, and in addition it normalizes W_M so $W_s = W_M \rtimes R$. Moreover, Kato defines the canonical subgroup

$$R(u) = \{w \in R \mid u^{n(w)} \text{ and } u \text{ are conjugate in } M^\vee\}.$$

Since $Z_{G^\vee}(x) = Z_{G^\vee}(s) \cap Z_{G^\vee}(u)$, we have that $Z_{M^\vee}(u) \lhd Z_{G^\vee}(x)$ and $Z_{G^\vee}(x)/Z_{M^\vee}(u) \cong R(u)$. In particular, we may regard $A_u^{M^\vee} := Z_{M^\vee}(u)/Z_{M^\vee}(u)^0$ as a normal subgroup of A_x, and we have the isomorphism $A_x/A_u^{M^\vee} \cong R(u)$.

Denote by $j : Z_{G^\vee}(s) \to R$ the canonical surjection and note that each $y \in Z_{G^\vee}(x) \leq Z_{G^\vee}(s)$ defines a map $H^i(\mathcal{B}_s^u) \to H^i(\mathcal{B}_s^u)$, which we also denote by y. Moreover, for each $w \in W_{M,\text{aff}}$, it follows from the definitions of the $W_{M,\text{aff}}$ that [Kat83, (6.1)]

$$y \cdot w = {}^{j(y)}w \cdot y : H^i(\mathcal{B}_s^u) \longrightarrow H^i(\mathcal{B}_s^u).$$

With this in mind, we consider the induced surjection $j_0 : A_x \to R(u)$ and, since $R(u)$ acts on $W_{M,\text{aff}}$ by conjugation, we can construct the semidirect product $A_x \ltimes W_{M,\text{aff}}$. This group embeds in $A_x \times W_{\text{aff}}$ via the map $(a, w) \mapsto (a, j_0(a)w)$ (the image in fact lands inside $A_x \times W_{s,\text{aff}}$), and Kato shows that the A_x-action on $H^*(\mathcal{B}_s^u)$ extends to an $A_x \ltimes W_{M,\text{aff}}$-action. In particular, we have the following result.

Proposition 4.4 ([Kat83, Proposition 6.2])**.** *As $A_x \times W_{\text{aff}}$-modules, we have*

$$H^*(\mathcal{B}^x) \cong \text{Ind}_{A_x \ltimes W_{M,\text{aff}}}^{A_x \times W_{\text{aff}}} H^*(\mathcal{B}_s^u).$$

Let us now go back to representations of p-adic groups. Let $V_{(x,\phi)}$ be an Iwahori-spherical unipotent representation of G. Then $V_{(x,\phi)}^{\mathcal{I}}$ is a simple $\mathcal{H}(G, \mathcal{I}, \mathbf{1})$-module, while $V_{(x,\phi),q=1}^{\mathbf{1}}$ is a simple W_{aff}-module. We then have the following central result, due to Lusztig. We remark that up to this point we have not used the fact that G is of simply connected type, so this result holds more generally by swapping W_{aff} with an adequate subgroup of the extended affine Weyl group $\widetilde{W}$.

Theorem 4.5. [Lus95b] *Let $V_{x,\phi}$ be an Iwahori-spherical irreducible representation of G. After letting $q \to 1$ and then taking semisimplification, the simple $\mathcal{H}(G, I, \mathbf{1})$-module $V_{x,\phi}^{\mathbf{1}}$ becomes the W_{aff}-module*

$\varepsilon \otimes H^*(\mathcal{B}^x)^\phi$, *where* ε *is the sign representation of* W_{aff}.

Remark 4.6. An important remark is due at this point. The above theorem relates the structure of $V^{\mathbf{1}}_{(x,\phi),q=1}$ with the structure of $H^*(\mathcal{B}^x)^\phi$ as W_{aff}-modules. However, the former object is naturally p-adic, while the latter is complex analytic. The Weyl groups of G and $G^\vee$ are canonically isomorphic, but not equal if the group is not simply laced, since the isomorphism swaps short with long roots. It is a standard abuse of notation to denote both Weyl groups with the same symbol, but one needs to be careful during explicit calculations inside which Weyl group one is working. We are ultimately interested in the p-adic side, but most computations involving cohomology groups and the Springer correspondence take place in the complex side. In particular, the reflection s_0 inside W will be a **short** reflection, corresponding to the highest short root. At the very end, we then trace the representations back to the p-adic side by swapping long and short roots.

Let us discuss some consequences of Theorem 4.5. When $R(u) = \{1\}$, the semidirect product $A_x \ltimes W_{M,\mathrm{aff}}$ is simply the direct product $A_x \times W_{M,\mathrm{aff}}$, and we have that

$$H^*(\mathcal{B}^x)^\phi = \left(\mathrm{Ind}_{A_x \times W_{M,\mathrm{aff}}}^{A_x \times W_{\mathrm{aff}}} H^*(\mathcal{B}_s^u)\right)^\phi = \mathrm{Ind}_{W_{M,\mathrm{aff}}}^{W_{\mathrm{aff}}} H^*(\mathcal{B}_s^u)^\phi. \tag{15}$$

From (14) and Theorem 4.5, the computation of the parahoric restriction of $V_{(x,\phi)}$ is then reduced to the computation of the $\widetilde{W}_J$-module structure of $H^*(\mathcal{B}^x)^\phi$. The difficulty of this case comes from simultaneously controlling the action of the character lattice X on $H^*(\mathcal{B}_s^u)^\phi$. In [Ree00], Reeder observes that the canonical projection from W_{aff} to W is injective on each $\widetilde{W}_J$ with image W_J. If $\psi_J : W_J \to \widetilde{W}_J$ is the inverse of this map, then we can consider the pullback ψ_J^* of representations of $\widetilde{W}_J$ to representations of W_J. By (15) and Mackey theory, we have that

$$\psi_J^* \left[\left(\mathrm{Ind}_{\widetilde{W}_M}^{\widetilde{W}} H^*(\mathcal{B}_s^u)^\phi \right) |_{\widetilde{W}_J} \right] \cong \bigoplus_{w \in W_M \backslash W / W_J} \mathrm{Ind}_{W_{J,s^w}}^{W_J} \chi_{s^w}^J \otimes \left[H^*(\mathcal{B}_s^u)^\phi \right]^w, \tag{16}$$

where $W_{J,s^w} = W_M^w \cap W_J$ and $\chi_{s^w}^J = \chi_{s^w} \circ \chi_J : W_{J,s^w} \to \mathbb{C}^\times$. To use this formula we need an understanding of the groups W_{J,s^w} and the characters $\chi_{s^w}^J$. The groups $W_{J,t}$ have been studied in the literature, and they can be decomposed into

$$W_{J,t} = W_{J,t}^0 \rtimes R_{J,t},$$

where $W_{J,t}^0$ is a "large" reflection subgroup of W generated by the reflections in J that fix t, and $R_{J,t}$ is a "small" group that normalizes $W_{J,t}^0$. The character χ_t^J is trivial on $W_{J,t}^0$ and is determined by its restriction to $R_{J,t}$. As a result, the characters χ_t^J are trivial in many cases. We refer to [Ree00, §8] for a detailed discussion on the structure of these characters, where explicit formulas to compute them are provided. If all characters $\chi_{s^w}^J$ appearing in (16) are trivial, then computations are greatly simplified.

Lemma 4.7. *With the same notation as in* (16)*, if the characters* $\chi_{s^w}^J$ *are trivial for all* $w \in$

$W_s\backslash W/W_J$, *then*

$$\psi_J^*\left[\left(\mathrm{Ind}_{\widetilde{W}_s}^{\widetilde{W}} H^*(\mathcal{B}_s^u)^\phi\right)|_{\widetilde{W}_J}\right] \cong \left(\mathrm{Ind}_{W_s}^{W} H^*(\mathcal{B}_s^u)^\phi\right)|_{W_J}$$

Proof. This is an immediate consequence of Mackey's formula applied to the right hand side of (16). □

4.3 Non-Iwahori-spherical case and weight diagrams

Let us now fix a general unipotent representation $V = V_{(x,\phi)}$ of G with associated parahoric $K := K_I$ and cuspidal unipotent representation σ as in (11). Then V^σ is a simple module over the Hecke algebra $\mathcal{H}(G, K, \sigma) = \mathrm{End}_G(c\text{-}\mathrm{Ind}_K^G \sigma)$. The structure of these Hecke algebras have been widely studied in the literature (see [Lus95a, §1] for a detailed discussion). In particular, one can associate a canonical extended affine Weyl group $\tilde{W}(K, \sigma)$ parametrizing a basis $\{T_w, w \in \tilde{W}(K, \sigma)\}$ of the algebra $\mathcal{H}(G, K, \sigma)$ satisfying explicit relations. Importantly, the basis elements $\{T_s \mid s \in S_{\mathrm{aff}}(K, \sigma)\}$ for some choice of simple reflections $S_{\mathrm{aff}}(K, \sigma)$ in $\tilde{W}(K, \sigma)$ satisfy the quadratic relations

$$(T_s + 1)(T_s - q^{c(s)}) = 0,$$

where $c(s)$ is an explicit positive integer depending on the simple root corresponding to s. The structure of a modules over $\mathcal{H}(G, K, \sigma)$ can then be largely understood by studying the action of the basis elements T_s corresponding to the simple reflections.

On the other hand, the extended affine Weyl group $\tilde{W}(K, \sigma)$ can be decomposed as a semidirect product $W(K, \sigma) \ltimes X(K, \sigma)$, where $W(K, \sigma)$ can be realized as the Weyl group of an associated complex reductive group $\mathbf{G}$ and $X(K, \sigma)$ as the character lattice of a maximal torus $\mathbf{T}$ of $\mathbf{G}$. The subalgebra spanned by $\{T_w, w \in W(K, \sigma)\}$ is naturally isomorphic to the subalgebra $\mathcal{H}(K_{J_0}, K, \sigma)$ of functions supported on a hyperspecial maximal compact subgroup K_{J_0} of G. We then have an explicit isomorphism ([Sol21, Theorem 1.10], [Lus89, §3])

$$\mathcal{H}(G, K, \sigma) \cong \mathcal{H}(K_{J_0}, K, \sigma) \,\tilde{\otimes}\, \mathcal{O}[\mathbf{T}], \tag{17}$$

where $\tilde{\otimes}$ is a twisted tensor product of $\mathbb{C}$-vector space with a twisted multiplication given by

$$(T_s - q^{c(s)})f - f^s(T_s - q^{c(s)}) = (f^s - f)\zeta_s \in \mathcal{O}[\mathbf{T}]$$

where $\zeta_s \in (f^s - f)^{-1}\mathcal{O}[\mathbf{T}]$ is a rational function on $\mathbf{T}$.

Example 4.8. Consider the case of $G = F_4(F)$, K a parahoric subgroup of type B_2 and σ the unique cuspidal unipotent representation of $B_2(\mathbb{F}_q)$. Then $\tilde{W}(K, \sigma)$ is the affine Weyl group of type C_2 and the Hecke algebra $\mathcal{H}(G, K, \sigma)$ is generated by the elements $T_{s_0}, T_{s_1}, T_{s_2}$ satisfying the quadratic relations

$$(T_{s_0} + 1)(T_{s_0} - q) = 0, \quad (T_{s_i} + 1)(T_{s_i} - q^3) = 0, \quad \text{for } i = 1, 2.$$

Moreover, $\mathbf{G} = \mathrm{SO}_5(\mathbb{C})$ and $\mathbf{T} = (\mathbb{C}^\times)^2$, with simple roots

$$e_{\alpha_1}(t_1, t_2) = t_1/t_2, \quad e_{\alpha_2}(t_1, t_2) = t_2,$$
$$\zeta_{s_1} = \frac{q^3 - e_{\alpha_1}}{1 - e_{\alpha_1}}, \quad \zeta_{s_2} = \frac{(q^2 - e_{\alpha_2})(q + e_{\alpha_2})}{1 - e_{2\alpha_2}}.$$

The theory of weight diagrams developed by Reeder in [Ree97] studies first the restriction of the module V^σ to the large subalgebra $\mathcal{O}[\mathbf{T}]$ and then the action of the basis elements T_s for $s \in W(K, \sigma)$ a simple reflection on the resulting decomposition. Since $\mathcal{O}[\mathbf{T}]$ is a commutative algebra, there is a decomposition

$$V^\sigma|_{\mathcal{O}[\mathbf{T}]} = \bigoplus_{\tau \in \mathbf{T}} V^\sigma(\tau),$$

where $V^\sigma(\tau)$ consists of those vectors in V^σ killed by some power of the maximal ideal of $\mathcal{O}[\mathbf{T}]$ corresponding to τ. From this decomposition, we can determine the action of the basis elements $W(K, \sigma)$ on the space $V^\sigma_{q=1}$, which immediately gives the parahoric restriction of V with respect to the maximal hyperspecial parahoric subgroup K_{J_0}. See [Ree00, §9.5] for the explicit formulas. To obtain the parahoric restriction with respect to other parahoric subgroups, one needs to understand the action of the remaining simple reflection $s_0 \in \tilde{W}(K, \sigma)$. This can be done by tracing back through the isomorphism $\mathcal{H}(G, K, \sigma) \cong \mathcal{H}(K_{J_0}, K, \sigma) \tilde{\otimes} \mathcal{O}[\mathbf{T}]$ and using the explicit formula for the twisted multiplication. See Example 4.8 for an explicit calculation. The remaining cases are omitted and we provide the final results without proof.

4.4 Action of the fundamental group

Let G_{ad} be the adjoint group of the simply connected group G and let $\eta : G \to G_{\mathrm{ad}}$ be the canonical map, which is in general not surjective. The standard adjoint action of G_{ad} on G induces a natural action on $\mathrm{Irr}_{un}(G)$ given by $(\mathrm{Ad}(g)^*\pi)(h) = \pi(g^{-1} \cdot h \cdot g)$ for $g \in G_{\mathrm{ad}}$ and $h \in G$. Of course, this action factors through $G_{\mathrm{ad}}/G := G_{\mathrm{ad}}/\eta(G)$, but is generally non-trivial since η is not surjective and G_{ad}/G acts on G by outer automorphisms. To discuss this action further, let T be a fixed maximal torus of G and let $T_{\mathrm{ad}} := T/Z(G)$ (Does this work without mentioning algebraic groups and F-points?) The extended affine Weyl group $\widetilde{W} := N_{G_{\mathrm{ad}}}(T_{\mathrm{ad}})(F)/T_{\mathrm{ad}}(\mathcal{O}_F)$ decomposes as $\widetilde{W} = W_{\mathrm{aff}} \rtimes \Omega$, where W_{aff} is the affine Weyl group acting simply transitively on the alcoves of $\mathcal{A}(G_{\mathrm{ad}}, T_{\mathrm{ad}}, F)$ and Ω is the fundamental alcove stabilizer. As such, lifts of Ω to G_{ad} permute the simple affine roots and moreover there is a short exact sequence

$$1 \longrightarrow G \xrightarrow{\eta} G_{\mathrm{ad}} \longrightarrow \Omega \longrightarrow 1,$$

where the second map is induced by the action of G_{ad} on the G-orbits of hyperspecial vertices on the Bruhat–Tits building $\mathcal{B}(G, F)$. In particular, for any $J \subsetneq S_{\mathrm{aff}}$ and any $g \in G_{\mathrm{ad}}$ corresponding to some

$\omega \in \Omega$ (that is, gK_Jg^{-1} and $K_{\omega(J)}$ are G-conjugate), we have that

$$\mathrm{res}_{\mathrm{un}}^{K_{\omega(J)}} \mathrm{Ad}(g)^*\pi \cong \mathrm{res}_{\mathrm{un}}^{K_J} \pi, \tag{18}$$

via the natural G_{ad}-conjugation inducing $K_J \cong K_{\omega(J)}$.

On the other hand, there is also a natural action of the group G_{ad}/G on the set of enhanced L-parameters constructed by Solleveld in [Sol20, §2] that preserves the set of unramified L-parameters. To discuss it, let $x = su$ be the Jordan decomposition of $x \in G^\vee$ and let $\phi \in \widehat{A_x}$ be an enhancement. Via Galois cohomology methods, Solleveld constructs a natural homomorphism (see [Sol20, §2.1])

$$\tau_x : G_{\mathrm{ad}}/G \longrightarrow \mathrm{Hom}(Z_{G^\vee}(s), \mathbb{C}^\times) \tag{19}$$

satisfying that $Z_{G^\vee}(s)^0 \subseteq \ker \tau_x(g)$ for all $g \in G_{\mathrm{ad}}$ ([Sol20, Lemma 2.1]). We can then restrict each $\tau_x(g)$ to $Z_{G^\vee}(x) \subseteq Z_{G^\vee}(s)$, which are trivial on $Z_{G^\vee}(x)^0$. Thus we obtain a natural homomorphism $\overline{\tau}_x : G_{\mathrm{ad}}/G \to \mathrm{Hom}(A_x, \mathbb{C}^\times)$ satisfying, by [Sol20, Theorem 2.10],

$$\mathrm{Ad}(g)^*\pi(x, \phi) = \pi(x, \phi \otimes \tau_x(g)^{-1}). \tag{20}$$

for any $\phi \in \widehat{A_x}$. Let us discuss the image of $\overline{\tau}_x$ in greater detail. Consider the dual map $\eta^\vee : G_{\mathrm{sc}}^\vee \to G^\vee$ and let $\tilde{u}$ be the unique unipotent lift of u in the simply connected cover $G_{\mathrm{sc}}^\vee$ of $G^\vee$ and fix any lift $\tilde{s}$ of s in $G_{\mathrm{sc}}^\vee$. Then $\tilde{x} = \tilde{s}\tilde{u}$ is a lift of x in $G_{\mathrm{sc}}^\vee$. Since $\tilde{u}$ is the unique unipotent lift of u, the canonical map $Z_{G_{\mathrm{sc}}^\vee}(\tilde{u}) \to Z_{G^\vee}(u)$ is surjective. While the map $Z_{G_{\mathrm{sc}}^\vee}(\tilde{s}) \to M^\vee := Z_{G^\vee}(s)^0$ is well-defined and surjective, the group $Z_{G^\vee}(s)$ may be disconnected, with component group isomorphic to R. Consider now the composition

$$Z_{G_{\mathrm{sc}}^\vee}(\tilde{x}) = Z_{Z_{G_{\mathrm{sc}}^\vee}(\tilde{s})}(\tilde{u}) \twoheadrightarrow Z_{M^\vee}(u) \hookrightarrow Z_{Z_{G^\vee}(s)}(u) = Z_{G^\vee}(x).$$

The first map is surjective (follows from the above observation) while the second map is clearly injective. Since $Z_{G^\vee}(s)/M^\vee \cong R$, we have that $Z_{G^\vee}(x)/Z_{M^\vee}(u) \cong R(u)$ by definition (recall here what R and $R(u)$ are). Recall that $A_{\tilde{x}}$ and A_x are the component groups of $Z_{G_{\mathrm{sc}}^\vee}(\tilde{x})$ and $Z_{G^\vee}(x)$, respectively, and let A_x^0 be the component group of $Z_{M^\vee}(u)$. It then follows that $A_x \cong A_x^0 \rtimes R(u)$, and we have the short exact sequence

$$1 \to Z/(Z \cap Z_{G_{\mathrm{sc}}^\vee}(\tilde{x})^0) \to A_{\tilde{x}} \to A_x^0 \to 1, \tag{21}$$

which implies that there is a bijection $\phi \leftrightarrow \tilde{\phi}$ between irreducible representations of A_x^0 and irreducible representations of $A_{\tilde{x}}$ trivial on the centre Z. Since all $\tau_x(g)$ are trivial on $Z_{G^\vee}(s)^0$, it follows that all $\overline{\tau}_x(g)$ are trivial on A_x^0 and therefore they all arise as inflations of characters of $R(u)$. In other words, the homomorphism $\overline{\tau}_x(g)$ factors through

$$\overline{\tau}_x(g) : G_{\mathrm{ad}}/G \longrightarrow \mathrm{Hom}(R(u), \mathbb{C}^\times) \xrightarrow{\inf} \mathrm{Hom}(A_x, \mathbb{C}^\times).$$

The following result fully determines the image of $\overline{\tau}_x(g)$.

Lemma 4.9. *The map $\overline{\tau}_x : G_{\mathrm{ad}}/G \to \mathrm{Hom}(R(u), \mathbb{C}^\times)$ is surjective.*

Proof. Given the Langlands parameter (x, ϕ) of G, we let $\tilde{x} \in G_{\mathrm{sc}}^\vee$ be as above and $\tilde{\phi}$ be the character of $A_{\tilde{x}}$ trivial on $Z(G_{\mathrm{sc}}^\vee)$ obtained by restricting ϕ to A_x^0 and the pulling back to $A_{\tilde{x}}$ via (21). Then $(\tilde{x}, \tilde{\phi})$ is naturally a Langlands parameter for G_{ad}. By [Sol20, Lemma 7.3], the pullback

$$\eta^*\pi(\tilde{x}, \tilde{\phi}) = \bigoplus_{\chi \in \widehat{A_x}} \mathrm{Hom}_{A_{\tilde{x}}}(\tilde{\phi}, \tilde{\chi}) \otimes \pi(x, \chi) = \bigoplus_{\rho \in \widehat{R(u)}} \pi(x, \phi \otimes \rho).$$

By standard Clifford theory, we know that G_{ad} must act transitively on the irreducible components of the right hand side, thus completing the proof. □

Since the p-adic group is simply-connected, maximal parahoric subgroups of G are in bijection with the maximal proper subsets of S_{aff}. To prove Conjecture 1.1 for G, one needs to verify the commutativity of the diagram for each $J \subsetneq S_{\mathrm{aff}}$. The above discussion allows us to verify this for a single $J \subsetneq S_{\mathrm{aff}}$ in each Ω orbit, as the next result shows.

Proposition 4.10. *Let G be a simple, split p-adic group of simply connected type. Let $J \subsetneq S_{\mathrm{aff}}$ and assume that Conjecture 1.1 holds for parahoric restrictions with respect to K_J. Then, for any $\omega \in \Omega$, Conjecture 1.1 also holds for parahoric restrictions with respect to $K_{\omega(J)}$.*

Proof. Let $u \in G^\vee$ unipotent and $(s, h) \in \mathcal{Y}(\Gamma_u)$ an elliptic pair. By assumption, we know that

$$\mathrm{FT}_{\mathrm{un}}^{\overline{K}_J} \circ \mathrm{res}_{\mathrm{un}}^{K_J} \Pi(u, s, h) = \zeta_{u,s,h} \mathrm{res}_{\mathrm{un}}^{K_J} \Pi(u, h, s) \text{ for some root of unity } \zeta_{u,s,h}. \tag{22}$$

A simple calculation then shows that

$$\Pi(u, s, h) = \sum_{\phi \in \widehat{A_{us}}} \phi(h)\pi(us, \phi) = \frac{1}{|R(u)|} \sum_{\phi \in \widehat{A_{us}}} \phi(h) \sum_{\chi \in \widehat{R(u)}} \chi(h)\pi(us, \phi \otimes \chi),$$

and therefore for any $g \in G_{\mathrm{ad}}$, (20)

$$\mathrm{Ad}(g)^*\Pi(u, s, h) = \frac{1}{|R(u)|} \sum_{\phi \in \widehat{A_{us}}} \phi(h) \sum_{\chi \in \widehat{R(u)}} \chi(h)\pi(us, \phi \otimes \chi\overline{\tau}_{us}(g)^{-1}) = \tag{23}$$

$$\frac{\overline{\tau}_{us}(g)(h)}{|R(u)|} \sum_{\phi \in \widehat{A_{us}}} \phi(h) \sum_{\chi \in \widehat{R(u)}} \chi(h)\overline{\tau}_{us}(g)(h)^{-1}\pi(us, \phi \otimes \chi\overline{\tau}_{us}(g)^{-1}) = \overline{\tau}_{us}(g)(h)\Pi(u, s, h), \tag{24}$$

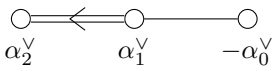
where $\overline{\tau}_{us}(g)(h)$ is a root of unity. Finally, it follows from (18) that

$$\mathrm{res}_{\mathrm{un}}^{K_{\omega(J)}} \mathrm{Ad}(g)^*\Pi(u, s, h) \cong \mathrm{res}_{\mathrm{un}}^{K_J} \Pi(u, s, h) \tag{25}$$

as $\overline{K}_J \cong \overline{K}_{\omega(J)}$-modules. The result now follows by combining (22) (23) and (25). □

4.5 Parahoric restriction for type G_2

We end this section with the explicit calculation of the parahoric restrictions for some representations $\pi(x,\phi)$ of the p-adic group $G = G_2(F)$, highlighting the methods described above with a concrete example. This group is both simply connected and adjoint, with dual group $G^\vee = G_2(\mathbb{C})$. Let α_1, α_2 be a choice of simple roots with α_1 short and let $\alpha_0 = 2\alpha_1 + \alpha_2,\ \alpha^0 = 3\alpha_1 + 2\alpha_2$ be the highest short and long root, respectively. The affine Dynkin diagram of $G_2(F)$ is



Thus $G_2(F)$ has 3 classes of maximal parahoric subgroups K_0, K_1, K_2 corresponding to the subsets of affine simple roots $J_0 = \{\alpha_1^\vee, \alpha_2^\vee\}, J_1 = \{-\alpha_0^\vee, \alpha_2^\vee\}$ and $J_2 = \{-\alpha_0^\vee, \alpha_1^\vee\}$ and with reductive types G_2, $A_1 \times \tilde{A}_1$ and A_2, respectively. We now compute these parahoric restrictions for all representations $\pi(us,\phi)$ of $G = G_2(F)$, where u is a subregular unipotent element of $G_2(\mathbb{C})$, labelled by $G_2(a_1)$. This is an interesting case since u is a distinguished unipotent element and $\Gamma_u = A_u \cong S_3$. There are 8 such representations, given by

$$\pi(G_2(a_1), 1, \phi), \phi \in \{\mathbf{1}, \varepsilon, r\}, \quad \pi(G_2(a_1), g_2, \phi), \phi \in \{\mathbf{1}, \varepsilon\}, \quad \pi(G_2(a_1), g_3, \phi), \phi \in \{\mathbf{1}, \theta, \theta^2\}.$$

Lusztig's arithmetic-geometric correspondence directly implies that 4 of them are unipotent supercuspidal representations of $G_2(F)$ given by

$$\pi(G_2(a_1), 1, \epsilon) = c\text{-Ind}_{K_0}^G G_2[1], \quad \pi(G_2(a_1), g_2, \epsilon) = c\text{-Ind}_{K_0}^G G_2[-1] \quad \text{and}$$
$$\pi(G_2(a_1), g_3, \theta^k) = c\text{-Ind}_{K_0}^G G_2[\theta^k], k = 1, 2,$$

while the remaining 4 are Iwahori-spherical. For the supercuspidal representations, their restrictions can be easily computed. By Frobenius reciprocity, their parahoric restrictions to K_0 are irreducible, while the remaining parahoric restrictions are 0.

To compute parahoric restriction of Iwahori-spherical representations, we recall that we work inside *complex* Weyl group W and consequently s_0 is as a short reflection along α_0, while s^0 is the long reflection along α^0. The unipotent element $u \in G^\vee = G_2(\mathbb{C})$ is subregular, so by Springer theory

$$H^*(\mathcal{B}^u)^{\mathbf{1}} = H^0(\mathcal{B}^u)^{\mathbf{1}} + H^2(\mathcal{B}^u)^{\mathbf{1}} = \phi_{1,0} + \phi_{2,1}, \quad \text{and} \quad H^*(\mathcal{B}^u)^r = H^0(\mathcal{B}^u)^r + H^2(\mathcal{B}^u)^r = \phi'_{1,3}$$

as W-modules. On the other hand, u is a regular unipotent element in $G_{g_2}^\vee = \mathrm{GL}_2(\mathbb{C})$ and $G_{g_3}^\vee = \mathrm{SL}_3(\mathbb{C})$, so $H^0(\mathcal{B}_{g_2}^u)$ and $H^0(\mathcal{B}_{g_3}^u)$ afford the trivial representations of $W_{g_2} = \langle s^0, s_1\rangle$ and $W_{g_3} = \langle s^0, s_2\rangle$ respectively, and

$$\mathrm{Ind}_{W_{g_2}}^W \mathbf{1} = \phi_{1,0} + \phi''_{1,3} \quad \text{and} \quad \mathrm{Ind}_{W_{g_3}}^W \mathbf{1} = \phi_{1,0} + \phi_{2,2}.$$

To obtain the parahoric restrictions to K_0, we note that all characters $\chi_t^{J_0}$ are trivial so it remains to twist by $\varepsilon = \phi_{1,6}$ (swapping $\phi_{1,0} \leftrightarrow \phi_{1,6}$ and $\phi'_{1,3} \leftrightarrow \phi''_{1,3}$) and then by the outer automorphism of W exchanging short and long roots (swapping $\phi'_{1,3} \leftrightarrow \phi''_{1,3}$ only).

To compute the parahoric restrictions with respect to K_1 and K_2 in Table 2 we compute the remaining characters χ_t^J. Using their properties above, one can check that all characters are trivial except for the pairs $(s, J) = (g_3, J_2)$ and $(s, J) = (g_2, J_1)$. When the characters are trivial, Lemma 4.7 and Alvis' tables immediately yield the results. The last two cases must be dealt with separately.

- **Restriction of $\pi(G_2(a_1), g_3, 1)$ to K_{J_2}.** In this case, $W_{J_2} = \langle s_0, s_1 \rangle$ (where s_0 is the reflection along the highest short root) and $W_{g_3} = \langle s^0, s_2 \rangle$ (where s^0 is the reflection along the highest root). Then $W_{g_3} \backslash W / W_{J_2} = \{1\}$, $W_{J_2, g_3} \cong C_3$ and $\chi_{g_3}^{J_2}$ is a primitive character. Thus,

$$\mathrm{Ind}_{W_{J_2, g_3}}^{W_{J_2}} \chi_{g_3}^{J_2} = r.$$

- **Restriction of $\pi(G_2(a_1), g_2, 1)$ to K_{J_1}.** In this case, $W_{J_1} = \langle s_0, s_2 \rangle$ and $W_{g_2} = \langle s^0, s_1 \rangle$, so $W_{g_2} \backslash W / W_{J_1} = \{1, w\}$ has two elements. Then, $W_{J_1, g_2} \cong C_2$ with $\chi_{g_2}^{J_1}$ non-trivial character and $W_{J_1, g_2^w} = W_{J_1}$ with $\chi_{g_2^w}^{J_1}$ the trivial character. Thus,

$$\mathrm{Ind}_{W_{J_1, g_2}}^{W_{J_1}} \chi_{g_2}^{J_1} + \mathrm{Ind}_{W_{J_1, g_2^w}}^{W_{J_1}} \chi_{g_2^w}^{J_1} = \varepsilon \boxtimes \mathbf{1} + \mathbf{1} \boxtimes \varepsilon + \mathbf{1} \boxtimes \mathbf{1}.$$

Twisting by the sign character yields the desired result. The proof of Conjecture 1.1 is now just a matter of putting all the pieces together. The parahoric restrictions of the virtual characters $\Pi(u, s, h)$ can be computed by linearity from the parahoric restrictions of the representations $\pi(u, s, \phi)$, which are given in Table 2.

	$K_0 \longrightarrow G_2$	$K_1 \longrightarrow A_1 \times \tilde{A}_1$	$K_2 \longrightarrow A_2$
$\Pi(G_2, 1, 1)$	$\phi_{1,6}$	$\epsilon \boxtimes \epsilon$	ϵ
$\Pi(G_2(a_1), 1, 1)$	$\phi_{2,1} + G_2[1] + 2\phi'_{1,3} + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + 3\epsilon \boxtimes \mathbf{1} + \epsilon \boxtimes \epsilon$	$3\epsilon + r$
$\Pi(G_2(a_1), 1, g_2)$	$\phi_{2,1} - G_2[1] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \mathbf{1} + \epsilon \boxtimes \epsilon$	$\epsilon + r$
$\Pi(G_2(a_1), g_2, 1)$	$\phi_{2,1} + \phi_{2,2} + G_2[-1] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \mathbf{1} + \epsilon \boxtimes \epsilon$	$\epsilon + r$
$\Pi(G_2(a_1), 1, g_3)$	$\phi_{2,1} + G_2[1] - \phi'_{1,3} + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \epsilon$	r
$\Pi(G_2(a_1), g_3, 1)$	$\phi''_{1,3} + G_2[\theta] + G_2[\theta^2] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \epsilon$	r
$\Pi(G_2(a_1), g_2, g_2)$	$\phi_{2,1} + \phi_{2,2} - G_2[-1] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \mathbf{1} + \epsilon \boxtimes \epsilon$	$\epsilon + r$
$\Pi(G_2(a_1), g_3, g_3)$	$\phi''_{1,3} + \theta G_2[\theta] + \theta^2 G_2[\theta^2] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \epsilon$	r
$\Pi(G_2(a_1), g_3, g_3^{-1})$	$\phi''_{1,3} + \theta^2 G_2[\theta] + \theta G_2[\theta^2] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \epsilon$	r

Table 2: Restrictions of G_2-representations $\pi(G_2(a_1), s, \phi)$

Finally, Lusztig's nonabelian Fourier transform acts trivially on every representation, except for the $G_2(\mathbb{F}_q)$ family

$$\{\phi_{2,1}, \phi'_{1,3}, G_2[1], \phi_{2,2}, G_2[-1], \phi''_{1,3}, G_2[\theta], G_2[\theta^2]\},$$

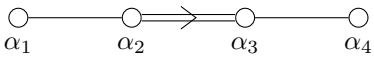
where the action is given by the 8×8 matrix

$$\begin{pmatrix}
\frac{1}{6} & \frac{1}{3} & \frac{1}{6} & \frac{1}{2} & \frac{1}{2} & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\
\frac{1}{3} & \frac{2}{3} & \frac{1}{3} & 0 & 0 & -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\
\frac{1}{6} & \frac{1}{3} & \frac{1}{6} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\
\frac{1}{2} & 0 & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 \\
\frac{1}{2} & 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\
\frac{1}{3} & -\frac{1}{3} & \frac{1}{3} & 0 & 0 & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\
\frac{1}{3} & -\frac{1}{3} & \frac{1}{3} & 0 & 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\
\frac{1}{3} & -\frac{1}{3} & \frac{1}{3} & 0 & 0 & -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3}
\end{pmatrix}$$

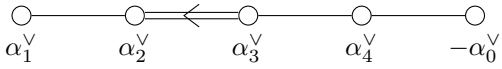
It is now a routine calculation to see that the square in Conjecture 1.1 commutes.

5 The p-adic group of type F_4

We now verify Conjecture 1.1 for the split, simply connected p-adic group $G = F_4(F)$. Since the fundamental group Ω is trivial, G is also adjoint. Its dual group is $G^\vee = F_4(\mathbb{C})$, and we let $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ be the simple roots of $G^\vee$, with Dynkin diagram



We denote by α_0 and α^0 the highest short root and highest long root of the root system, respectively, so the simple affine roots of G are $S_{\text{aff}} = \{-\check{\alpha}_0, \check{\alpha}_4, \check{\alpha}_3, \check{\alpha}_2, \check{\alpha}_1\}$ with affine Dynkin diagram



The group G has five maximal open compact subgroups $\{K_0, K_4, K_3, K_2, K_1\}$ up to conjugacy, where K_i corresponds to $J_i := S_{\text{aff}} \setminus \{\alpha_i^\vee\}$ for $0 \leq i \leq 4$. The reductive quotients $\overline{K}_0, \overline{K}_4, \overline{K}_3, \overline{K}_2$ and $\overline{K}_1$ are finite reductive groups of type $F_4, A_1 \times C_3, A_2 \times \tilde{A}_2, A_3 \times \tilde{A}_1$ and B_4, respectively. In this section, we prove the following.

Theorem 5.1. *Let G be a simple p-adic group of type F_4. Then the diagram*

$$\begin{array}{ccc} R_{\text{un,ell}}(G) & \xrightarrow{\mathrm{FT}^\vee_{\text{ell}}} & R_{\text{un,ell}}(G) \\ \downarrow{\scriptstyle \mathrm{res}^{\text{cpt}}_{\text{un}}} & & \downarrow{\scriptstyle \mathrm{res}^{\text{cpt}}_{\text{un}}} \\ \bigoplus_{i=0}^4 R_{\text{un}}(\overline{K}_i) & \xrightarrow{(\mathrm{FT}^{K_i})_i} & \bigoplus_{i=0}^4 R_{\text{un}}(\overline{K}_i), \end{array}$$

commutes.

We remark that the above result was already proven by Ciubotaru, in an unpublished paper [Ciu20] for the reduction to the hyperspecial parahoric subgroup K_0. We complete the proof for the remaining 4 parahoric subgroups.

Firstly, we compute the basis $\{\Pi(u, s, h) \mid u \in G^\vee \text{ unipotent}, (s, h) \in \Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{\text{ell}}\}$ of the space $R_{\text{un,ell}}(G)$ (see Section 3.1 for the definitions). The classification of all elliptic pairs of $F_4(\mathbb{C})$ has been previously computed in the literature, and the list is given in Table 3. For each unipotent $u \in F_4(\mathbb{C})$, we also give the unipotent $F_4(\mathbb{F}_q)$-family $\mathcal{F}_u$ whose special element is the W-representation afforded by $\varepsilon \otimes H^{\text{top}}(\mathcal{B}^u)^{\mathbf{1}}$ under the Springer correspondence. Then $\Gamma_{\mathcal{F}_u}$ is the finite group canonically associated to the family $\mathcal{F}_u$.

For example, when u has type $F_4(a_3)$, $\Gamma_u \cong S_4$ and there are 21 virtual characters $\Pi(u, s, h)$ (below

Unipotent	Γ_u^0	A_u	$\lvert\Gamma_u\backslash\mathcal{Y}(\Gamma_u)_{\mathrm{ell}}\rvert$	$\mathcal{F}_u$	$\Gamma_{\mathcal{F}_u}$
F_4	1	1	1	$\phi_{1,24}$	1
$F_4(a_1)$	1	S_2	4	$\phi_{4,13}$	C_2
$F_4(a_2)$	1	S_2	4	$\phi_{9,10}$	1
B_3	PGL(2)	1	1	$\phi''_{8,9}$	1
$F_4(a_3)$	1	S_4	21	$\phi_{12,4}$	S_4

Table 3: Elliptic pairs for $F_4(\mathbb{C})$

$\tau = g_2 \cdot g_2'$ and γ is conjugate to g_2' in S_4 but not in $Z_{S_4}(g_2') = D_8$):

$$\Pi(F_4(a_3),1,1) = [F_4(a_3),4] + 3[F_4(a_3),31] + 3[F_4(a_3),211] + [F_4(a_3),1^4] + 2[F_4(a_3),22];$$
$$\Pi(F_4(a_3),1,g_2) = [F_4(a_3),4] + [F_4(a_3),31] - [F_4(a_3),211] - [F_4(a_3),1^4];$$
$$\Pi(F_4(a_3),1,g_2') = [F_4(a_3),4] - [F_4(a_3),31] - [F_4(a_3),211] + [F_4(a_3),1^4] + 2[F_4(a_3),22];$$
$$\Pi(F_4(a_3),1,g_3) = [F_4(a_3),4] + [F_4(a_3),1^4] - [F_4(a_3),22];$$
$$\Pi(F_4(a_3),1,g_4) = [F_4(a_3),4] - [F_4(a_3),31] + [F_4(a_3),211] - [F_4(a_3),1^4];$$

$$\Pi(F_4(a_3),g_2,1) = [C(42)A_1,++] + [C(42)A_1,+-] + [C(42)A_1,-+] + [C(42)A_1,--];$$
$$\Pi(F_4(a_3),g_2,g_2) = [C(42)A_1,++] - [C(42)A_1,+-] + [C(42)A_1,-+] - [C(42)A_1,--];$$
$$\Pi(F_4(a_3),g_2,\tau) = [C(42)A_1,++] + [C(42)A_1,+-] - [C(42)A_1,-+] - [C(42)A_1,--];$$
$$\Pi(F_4(a_3),g_2,g_2') = [C(42)A_1,++] - [C(42)A_1,+-] - [C(42)A_1,-+] + [C(42)A_1,--];$$

$$\Pi(F_4(a_3),g_2',1) = [B_4(531),\mathbf{1}] + 2[B_4(531),r] + [B_4(531),\varepsilon'] + [B_4(531),\varepsilon''] + [B_4(531),\varepsilon];$$
$$\Pi(F_4(a_3),g_2',g_2') = [B_4(531),\mathbf{1}] - [B_4(531),\varepsilon'] + [B_4(531),\varepsilon''] - [B_4(531),\varepsilon];$$
$$\Pi(F_4(a_3),g_2',g_2'') = [B_4(531),\mathbf{1}] - 2[B_4(531),r] + [B_4(531),\varepsilon'] + [B_4(531),\varepsilon''] + [B_4(531),\varepsilon];$$
$$\Pi(F_4(a_3),g_2',g_4) = [B_4(531),\mathbf{1}] - [B_4(531),\varepsilon'] - [B_4(531),\varepsilon''] + [B_4(531),\varepsilon];$$
$$\Pi(F_4(a_3),g_2',\gamma) = [B_4(531),\mathbf{1}] + [B_4(531),\varepsilon'] - [B_4(531),\varepsilon''] - [B_4(531),\varepsilon];$$

$$\Pi(F_4(a_3),g_3,1) = [2A_2,\mathbf{1}] + [2A_2,\theta] + [2A_2,\theta^2];$$
$$\Pi(F_4(a_3),g_3,g_3) = [2A_2,\mathbf{1}] + \theta^2[2A_2,\theta] + \theta[2A_2,\theta^2];$$
$$\Pi(F_4(a_3),g_3,g_3^{-1}) = [2A_2,\mathbf{1}] + \theta[2A_2,\theta] + \theta^2[2A_2,\theta^2];$$

$$\Pi(F_4(a_3),g_4,1) = [A_3A_1,1] + [A_3A_1,i] + [A_3A_1,-1] + [A_3A_1,-i];$$
$$\Pi(F_4(a_3),g_4,g_4) = [A_3A_1,1] - i[A_3A_1,i] - [A_3A_1,-1] - i[A_3A_1,-i];$$
$$\Pi(F_4(a_3),g_4,g_2') = [A_3A_1,1] - [A_3A_1,i] + [A_3A_1,-1] - [A_3A_1,-i];$$
$$\Pi(F_4(a_3),g_4,g_4^{-1}) = [A_3A_1,1] + i[A_3A_1,i] - [A_3A_1,-1] - i[A_3A_1,-i];$$

These virtual characters span a 21-dimensional space inside the 31-dimensional space $R_{\mathrm{un,ell}}(G)$. All remaining 10 basis elements are easily computable. The proof of Theorem 5.1 relies on an explicit computation of the restriction map

$$\mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}} : R_{\mathrm{un,ell}}(G) \longrightarrow \bigoplus_{i=0}^{5} R_{\mathrm{un}}(\overline{K}_i).$$

Most of this work was completed by Reeder in [Ree00]. Nevertheless, his tables contain a few mistakes (systematically arising from the two representations $\pi(F_4(a_3), 1, 31)$ and $\pi(F_4(a_3), 1, 211)$) and do not include the representations whose unipotent label u has type B_3 (for this is not a distinguished unipotent element of $F_4(\mathbb{C})$). We tabulate these results at the end of the section, fixing the mistakes and adding the new restrictions.

Let us briefly discuss how the parahoric restrictions are computed. The unipotent representations of $F_4(F)$ fall into 9 unipotent blocks, together with their associated Hecke algebras. There are 7 supercuspidal blocks, one for each cuspidal unipotent representation of $F_4(\mathbb{F}_q)$. The corresponding Hecke algebras $\mathcal{H}(F_4(F), F_4(\mathcal{O}_F), \sigma)$ are all isomorphic to $\mathbb{C}$ so these blocks contain a unique unipotent representation. This information can be read from Lusztig's arithmetic-geometric correspondence tables in [Lus95a, pg. 577]. The Langlands parameters of these representations are

$$[F_4(a_3), 1^4],\ [C_3(42), --],\ [B_4(531), \varepsilon],\ [2A_2, \theta],\ [2A_2, \theta^2],\ [A_1A_3, i] \text{ and } [A_1A_3, -i],$$

where we are using Notation 3.6 for the Langlands parameters. The parahoric restrictions of these representations are straightforward to compute, for they are irreducible for the hyperspecial parahoric K_0 and vanish for the other ones.

The principal block contains all Iwahori-spherical representations, that is, all representations $\pi(us, \phi)$ such that $H(\mathcal{B}_s^u)^\phi \neq 0$. The corresponding Hecke algebra $\mathcal{H}(F_4(F), I, \mathbf{1})$ is the affine Hecke algebra of type F_4 with equal parameters. We compute the parahoric restrictions, using Springer theory, the Green tables of $F_4(\mathbb{F}_q)$ and the methods described in subsection 4.2.

Finally, there is an intermediate block whose Hecke algebra $\mathcal{H}(F_4(F), B_2, \sigma)$ is the affine Hecke algebra of type C_2 with unequal parameters that we introduced in Example 4.8. All Langlands parameters of the representations of interest in this block are *elliptic*, so it suffices to compute the square integrable representations of $\mathcal{H}(F_4(F), B_2, \sigma)$ to determine the desired parahoric restrictions. This can be achieved using the theory of weight diagrams briefly discussed in Section 4.3 (see also [Ciu20, §9]). The Langlands parameters of these representations are

$$[B_4, -],\ [C_3A_1, -],\ [C_3(42)A_1, -+],\ [B_4(531), r] \text{ and } [A_1A_3, -1].$$

We explicitly discuss the $[A_1A_3, -1]$ case in Example 5.5, and we tabulate the rest of the parahoric restrictions in Section 7. The proof of Theorem 5.1 follows immediately from Tables 7, 9, 11 and 14.

Let us finish this section by discussing a few examples in detail to illustrate the main ideas involved in the calculations.

5.1 Examples of parahoric restrictions

The first example is the computation of a parahoric restrictions with respect to the maximal hyperspecial parahoric K_0 using Springer Theory and Alvis' tables ([Alv05]).

Example 5.2. Consider the representation $[B_4(531), \mathbf{1}]$, with u of type $F_4(a_3)$ and $s \in \Gamma_u$ a product of two 2-cycles. The centralizer $Z_{G^\vee}(s)$ has type B_4 and $u \in Z_{G^\vee}(s)$ is subregular. Thus, the Springer fibre $\mathcal{B}_s^u$ is 2 dimensional, and by reading the Green functions we obtain

$$H^*(\mathcal{B}_s^u)^{\mathbf{1}} = [2,2] + [3,1] + [4,\cdot] \quad \text{as } W_s\text{-modules.}$$

Since all characters $\chi_{s^w}^J$ are trivial when $J = J_0$, Alvis' tables yield

$$\mathrm{Ind}_{W_s}^W [2,2] + [3,1] + [4,\cdot] = \phi_{9,2} + \phi''_{9,6} + \phi''_{8,3} + \phi_{4,1} + \phi''_{2,4} + \phi_{1,0}.$$

To obtain the parahoric restriction of the representation $[B_4(531), \mathbf{1}]$ with respect to K_0, we twist by ε and by the outer automorphism of W swapping long and short roots, so

$$[B_4(531), \mathbf{1}] \xrightarrow{\mathrm{res}_{\mathrm{un}}^{K_0}} \phi_{9,10} + \phi''_{9,6} + \phi''_{8,9} + \phi_{4,13} + \phi''_{2,16} + \phi_{1,24}.$$

Once the parahoric restrictions for the maximal hyperspecial have been calculated, we can turn our attention towards computing the rest. The first step is to determine when the characters χ_t^J are trivial, which happens often and it can be easily determined from [Ree00, Lemma 8.3]. Whenever this is the case, we can use Lemma 4.7 and Alvis' tables to compute the restriction directly. Let us discuss another interesting example.

Example 5.3. Consider the parahoric restriction of the Iwahori-spherical representation $[A_1A_3, \mathbf{1}]$, where u is of type $F_4(a_3)$ and $s \in \Gamma_u$ is a 4-cycle, with respect to the parahoric subgroup K_2. Note that $W_{J_2} = \langle s_0, s_4, s_3, s_1 \rangle \cong W(A_3) \times W(A_1)$. The semisimple part s can be chosen to be $s = \check{\alpha}_3(-1)\check{\alpha}_4(i)$ so that $W_s = \langle s_4, s_2, s_1, s^0 \rangle \cong W(A_1) \times W(A_3)$. In addition, u is a regular unipotent element in $Z_{G^\vee}(s)$, so $H^*(\mathcal{B}_s^u)^{\mathbf{1}} = \mathbf{1}$ as a W_s-module. Next,

$$W_s \backslash W / W_{J_2} = \{1, w\},$$

where $w \in W$ is explicitly computable with computer algebra programs such as GAP. In addition,

$$W_{J_2,s} = \langle s_4, s_1 \rangle \cong C_2 \times C_2, \quad \text{while} \quad W_{J_2,s^w} = \langle d \rangle \cong C_4,$$

for some $d \in W$ of order 4. Since $W_{J_2,s}$ is a reflection subgroup of W_{J_2}, the character $\chi_s^{J_2}$ is trivial. On the other hand, W_{J_2,s^w} is not a reflection subgroup of W_{J_2}, and $\chi_{s^w}^{J_2} = \chi$ is a primitive character of C_4. The induced representations are then given by

$$\mathrm{Ind}_{W_{J_2,s}}^{W_{J_2}} \mathbf{1} = [[4] + 2[311] + [22] + [211]] \boxtimes [2] \quad \text{and} \quad \mathrm{Ind}_{W_{J_2,s^w}}^{W_{J_2}} \chi = [[31] + [211]] \boxtimes [[2] + [11]].$$

Adding these and twisting by sign gives the required parahoric restriction.

Example 5.4. A more challenging example is the calculation of the parahoric restriction of $[2A_2, \mathbf{1}]$, with u of type $F_4(a_3)$ and $s \in \Gamma_u$ is a 3-cycle, with respect to K_3. This is the only case where a characters $\chi_{s^w}^{J_3}$ may be non-trivial. Note that $W_{J_3} = \langle s_0, s_4, s_2, s_1 \rangle \cong W(A_2) \times W(A_2)$. The semisimple part can be taken to be $s = \check{\alpha}_3(\xi_3)\check{\alpha}_4(\xi_3^2)$, where ξ_3 is a primitive third root of unity. In particular, $Z_{G^\vee}(s)$ is a pseudo-Levi subgroup of type $A_2 \times A_2$ generated by the affine simple roots $\{\alpha_4, \alpha_3, \alpha_1, -\alpha^0\}$ and $W_s = \langle s_4, s_3, s_1, s^0 \rangle$. In addition, the unipotent element u is regular in $Z_{G^\vee}(s)$, so $H^*(\mathcal{B}_s^u)^{\mathbf{1}} = \mathbf{1}$ as W_s-modules. Next,

$$W_s \backslash W / W_{J_3} = \{w_1 = 1, w_2, w_3, w_4, w_5\},$$

where $w_i \in W$ are representatives of double cosets with sizes $324, 324, 36, 36$ and 432, respectively. The subgroups

$$W_{J_3,s^{w_3}} = W_{J_3,s^{w_4}} = W_{J_3} \quad \text{and} \quad W_{J_3,s} \cong W_{J_3,s^{w_2}} \cong C_2 \times C_2$$

are all reflections subgroups of W_{J_3}, so the characters $\chi_{s^{w_i}}^{J_3}$ are trivial for $i = 1, 2, 3, 4$. The remaining coset is more interesting since

$$W_{J_3,s^{w_5}} = \langle d \rangle \cong C_3,$$

is no longer a reflection subgroup, and $\chi_{s^{w_4}}^{J_3}$ is a primitive character χ of C_3.

The induced representations are then given by

$$\mathrm{Ind}_{W_{J_3,s}}^{J_3} \mathbf{1} = \mathrm{Ind}_{W_{J_3,s^{w_2}}}^{J_3} \mathbf{1} = [3,3] + [3,21] + [21,3] + [21,21], \quad \mathrm{Ind}_{W_{J_3,s^{w_3}}}^{J_3} \mathbf{1} = \mathrm{Ind}_{W_{J_3,s^{w_4}}}^{J_3} \mathbf{1} = [3,3]$$
$$\text{and} \quad \mathrm{Ind}_{W_{J_3,s^{w_5}}}^{J_3} \chi = [21,3] + [3,21] + [21,21] + [21,1^3] + [1^3,21].$$

Adding these and twisting by sign gives the required parahoric restriction.

Finally, let us discuss the parahoric restrictions of the non-Iwahori-spherical representation $[A_1A_3, -1]$.

Example 5.5. Let V be the representation labelled by $[A_1A_3, -1]$, K a parahoric subgroup of type B_2 and let σ be the unique cuspidal unipotent representation of $B_2(\mathbb{F}_q)$. Since the types of the reductive quotients of K_2 and K_3 ($A_3 \times \tilde{A}_1$ and $A_2 \times \tilde{A}_2$, respectively) do not contain B_2 as a factor, the parahoric restrictions of V to K_2 and K_3 vanish.

On the other hand, V^σ is a square integrable representation of $\mathcal{H}(F_4(F), B_2, \sigma) \cong \mathcal{H}(K_{J_0}, K, \sigma) \tilde{\otimes} \mathcal{O}[\mathbf{T}]$,

whose restriction to $\mathcal{O}[\mathbf{T}]$ is

$$V^\sigma|_{\mathcal{O}[\mathbf{T}]} = V^\sigma(q^{-2}, -q^{-1}) \oplus V^\sigma(-q^{-1}, q^{-2}),$$

both subspaces being one-dimensional. By choosing a basis element in each subspace, the rational functions $e_{\alpha_1}, e_{\alpha_2} \in \mathcal{O}[\mathbf{T}]$ act on V^σ by $\mathrm{diag}(-q^{-1}, -q)$ and $\mathrm{diag}(-q^{-1}, q^{-2})$. Using the formula for the twisted multiplication in $\mathcal{H}(F_4(F), B_2, \sigma)$, the action of the generators T_1 and T_2 can be computed explicitly, and we obtain

$$T_{s_1} \longrightarrow \frac{1}{q+1}\begin{pmatrix} q^3-1 & q^3+q \\ q^4+1 & q^4-q \end{pmatrix} \quad \text{and} \quad T_{s_2} \longrightarrow \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Finally, one can trace back through the isomorphisms $\mathcal{H}(F_4(F), B_2, \sigma) \cong \mathcal{H}(K_{J_0}, K, \sigma) \tilde{\otimes} \mathcal{O}[\mathbf{T}]$ to obtain that $T_{\tilde{s}_0}$ acts by

$$T_{\tilde{s}_0} \longrightarrow \frac{1}{q^3+q^2}\begin{pmatrix} -q^6+q^3-q^2-1 & q^5+q^3-q^2-1 \\ -q^7+q^4-q^3+1 & q^6+q^4-q^3+1 \end{pmatrix}.$$

Thus, the action of $\tilde{W}(K, \sigma)$ on $V^\sigma_{q=1}$ is given by

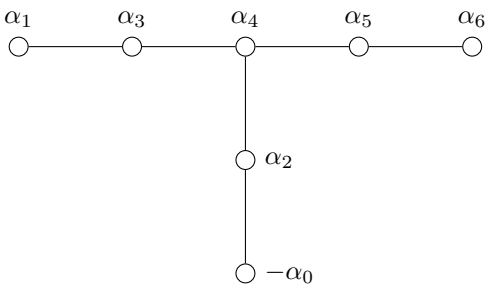
$$s_1 \longrightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad s_2 \longrightarrow \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{and} \quad \tilde{s}_0 \longrightarrow \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The parahoric restrictions are now obtained by restricting to reflection subgroups. Restriction to the reflection subgroup $\langle s_1, s_2\rangle \cong W(B_2)$ gives the parahoric restriction to K_0, restriction to $\langle s_1, \tilde{s}_0\rangle \cong W(B_2)$ gives the parahoric restriction to K_1 and restriction to $\langle s_2, \tilde{s}_0\rangle \cong W(A_1 \times A_1)$ gives the parahoric restriction to K_4. In each case, there is a bijection between representations of the Weyl group and unipotent representations of $\overline{K}_i$ in the series generated by σ, which we use implicitly. We refer the reader to [Car85, §10.11 and §13.9] Throughout, we use Lusztig's notation for the unipotent representations. We obtain

- Firstly, $V^\sigma|_{\langle s_1, s_2\rangle} = [11, \emptyset] + [\emptyset, 11]$, so $[A_1A_3, -1] \xrightarrow{\mathrm{res}^{K_0}_{\mathrm{un}}} B_2[\varepsilon'] + B_2[\varepsilon]$.
- Secondly, $V^\sigma|_{\langle s_1, \tilde{s}_0\rangle} = [1, 1]$, so $[A_1A_3, -1] \xrightarrow{\mathrm{res}^{K_1}_{\mathrm{un}}} ({}^{0\ 1}_{1}{}^{2\ 4})$.
- Finally, $V^\sigma|_{\langle s_2, \tilde{s}_0\rangle} = [11] \boxtimes [2] + [11] \boxtimes [11]$, so $[A_1A_3, -1] \xrightarrow{\mathrm{res}^{K_4}_{\mathrm{un}}} ({}^{0\ 1}_{1}{}^{2\ 3}) \boxtimes \mathbf{1} + ({}^{0\ 1}_{1}{}^{2\ 3}) \boxtimes \mathrm{St}$.

6 The p-adic group of type E_6

In this section, we let $G = E_6(F)_{\mathrm{sc}}$ be the simple p-adic of type E_6 of simply connected type. Its affine Dynkin diagram is



and its fundamental group is $\Omega \cong \mathbb{Z}/3\mathbb{Z}$. The group G has 7 conjugacy classes of open maximal open compact subgroups $K_i,\ i = 0, 1, \ldots, 6$, one for each $J_i = S_{\mathrm{aff}} \setminus \{\check{\alpha}_i\},\ i = 0, 1, \ldots, 6$. In this section, we prove

Theorem 6.1. *Let G be a simple p-adic group of type E_6 of simply connected type. Then the diagram*

$$\begin{array}{ccc} R_{\mathrm{un,ell}}(G) & \xrightarrow{\mathrm{FT}^{\vee}_{\mathrm{ell}}} & R_{\mathrm{un,ell}}(G) \\ \downarrow{\scriptstyle \mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}} & & \downarrow{\scriptstyle \mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}} \\ \bigoplus_{i=0}^{6} R_{\mathrm{un}}(\overline{K}_i) & \xrightarrow{(\mathrm{FT}^{K_i})_i} & \bigoplus_{i=0}^{6} R_{\mathrm{un}}(\overline{K}_i), \end{array}$$

commutes, up to roots of unity.

The classes parametrized by J_0, J_1, J_6 and J_2, J_3, J_5 are conjugate under the action of Ω, respectively, and in particular they have the same reductive type. By Proposition 4.10, it is therefore enough to verify the conjecture for the conjugacy classes of maximal open compact subgroups corresponding to J_0, J_2 and J_4.

Next, we classify the elliptic pairs in $G^{\vee} = E_6(\mathbb{C})_{\mathrm{ad}}$. This classification is well known in the literature for $E_6(\mathbb{C})_{\mathrm{sc}}$, but for $E_6(\mathbb{C})_{\mathrm{ad}}$ is slightly more involved. For most unipotent elements $u \in G^{\vee}$, the reductive part Γ_u of its centralizer is connected. When this is the case, there exists a precise description of the elliptic pairs in Γ_u.

Proposition 6.2. [ACR25, Proposition 8.7] *Suppose that Γ_u is connected. Then there is a bijection*

$$\Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{ell} \leftrightarrow W(\Gamma_u) \backslash \{(s, w) \mid s \in T \text{ is regular, } w \in W(\Gamma_u) \text{ is elliptic, } s \in T^w\},$$

where the map is obtained by conjugating an elliptic pair (s, h) so that s lies in a fixed torus T of Γ_u and h is the lift of some $w \in W(\Gamma_u)$ such that $s \in T^w$.

With this in mind we can give a precise description of the elliptic pairs inside $G^{\vee} = E_6(\mathbb{C})_{\mathrm{ad}}$. We use

the same notation as in Table 3.

Unipotent	Γ_u^0	A_u	$\lvert\Gamma_u\backslash\mathcal{Y}(\Gamma_u)_{\mathrm{ell}}\rvert$	$\mathcal{F}_u$	$\Gamma_{\mathcal{F}_u}$
E_6	1	1	1	$\phi_{1,36}$	1
$E_6(a_1)$	1	1	1	$\phi_{6,25}$	1
$E_6(a_3)$	1	S_2	4	$\phi_{30,15}$	C_2
D_4	PGL_3	1	2	$\phi_{24,12}$	1
$D_4(a_1)$	$(\mathbb{C}^\times)^2$	S_3	12	$\phi_{80,7}$	S_3
A_2	$(\mathrm{SL}_3\times\mathrm{SL}_3)/\mu_3$	1	2	$\phi_{30,3}$	C_2

Table 4: Elliptic pairs for $E_6(\mathbb{C})_{\mathrm{ad}}$

Let us briefly describe the elliptic pairs above explicitly. When $\Gamma_u^0 = \{1\}$ the description is straightforward. Next, consider the semisimple matrices

$$s_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \zeta_3 & 0 \\ 0 & 0 & \zeta_3^2 \end{pmatrix} \quad \text{and} \quad w_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \quad \text{in } \mathrm{SL}_3(\mathbb{C}).$$

We also let $s_{3,3} = (s_3, s_3)$ and $w_{3,3} = (w_3, w_3)$ be matrices in $\mathrm{SL}_3 \times \mathrm{SL}_3$. We abuse notation and denote their images in central quotients with the same symbols as necessary.

When u has type D_4, then $\Gamma_u\backslash\mathcal{Y}(\Gamma_u)_{\mathrm{ell}} = \{(s_3, w_3), (s_3, w_3^2)\}$ and $A_{us_3} = R(u) \cong C_3$, while if u has type A_2, then $\Gamma_u\backslash\mathcal{Y}(\Gamma_u)_{\mathrm{ell}} = \{(s_{3,3}, w_{3,3}), (s_{3,3}, w_{3,3}^2)\}$ and $A_{us_3} = R(u) \cong C_3$ as well. The remaining case when u has type $D_4(a_1)$ is the most interesting. Then $\Gamma_u = (\mathbb{C}^\times)^2 \rtimes S_3$, where the 3-cycle (123) in S_3 acts by $(a, b) \mapsto (b, \frac{1}{ab})$ and the 2-cycle (12) acts by $(a, b) \mapsto (\frac{1}{b}, \frac{1}{a})$. A lengthy computation then shows that Γ_u has 12 elliptic pairs up to conjugacy. To describe them, let

$$1 = ((1,1), e),\ z = ((\zeta_3, \zeta_3), e),\ g_3 = ((1,1), (123)) \in \Gamma_u.$$

Then the 12 elliptic pairs up to conjugacy are $\{(z^k, g_3), (g_3, z^k), (g_3, z^k g_3), (g_3, z^k g_3^{-1}) \mid k = 0, 1, 2\}$. The flip $(s, h) \mapsto (h, s)$ in the space of elliptic pairs preserves the pairs $(g_3, g_3^{\pm 1})$, while it flips $(z^k, g_3) \leftrightarrow (g_3, z^k)$ and $(g_3, zg_3^{\pm 1}) \leftrightarrow (g_3, z^2 g_3^{\pm 1})$. For future computations, we also note that $(z^2, g_3) \sim (z, g_3^{-1})$ and that

$$A_u = A_u^0 = S_3, \quad A_{uz} = R(u) = \langle g_3 \rangle \cong C_3 \quad \text{and} \quad A_{ug_3} = A_{ug_3}^0 \times R(u) = \langle g_3 \rangle \times \langle z \rangle \cong C_3 \times C_3$$

Thus, $R_{\mathrm{un,ell}}(G)$ is a 22-dimensional vector space, and the proof of 6.1 relies again on an explicit computation of the parahoric restriction maps

$$\mathrm{res}_{\mathrm{un}}^{K_i} : R_{\mathrm{un,ell}} \longrightarrow R_{\mathrm{un}}(\overline{K}_i), \quad \text{for } i = 0, 2, 4.$$

Finally, let us describe the unipotent blocks and the Langlands parameters for G. There are 8 unipotent blocks, 6 of which are supercuspidal, one for each pair (K_0, σ), where $K_0 \cong E_6(\mathcal{O}_F)$ is a hyperspecial

parahoric subgroup (there are three conjugacy classes in G) and $\sigma \in \{E_6[\theta], E_6[\theta^2]\}$ is a cuspidal unipotent representation of $E_6(\mathbb{F}_q)$. The Hecke algebras $\mathcal{H}(E_6(F), K_0, \sigma) \cong \mathbb{C}$ so the block contains a unique Langlands parameter, which can be read from Lusztig's arithmetic-geometric tables (when G is adjoint), combined with Solleveld's results ([Sol20, Lemma 7.3]).

Again, the principal block contains all representations $\pi(us, \phi)$ such that $H(\mathcal{B}_s^u)^\phi \neq 0$ and the corresponding Hecke algebra $\mathcal{H}(E_6(F), \mathcal{I}, \mathbf{1})$ is the affine Hecke algebra of type E_6 of equal parameters. The parahoric restrictions are computed using the methods described in Section 4.

Finally, there is one intermediate Hecke algebra $\mathcal{H}(E_6(F), D_4(\mathcal{O}_F), \sigma)$, where σ is the unique cuspidal unipotent representation of $D_4(\mathbb{F}_q)$. The only Langlands parameter we encounter corresponding to this Hecke algebra is $[A_1A_5, -]$, where u has type $E_6(a_3)$, s is the unique non-trivial semisimple element commuting with u (of order 2), and ϕ is non-trivial.

We conclude this section with an explicit expression of the virtual characters $\Pi(u, s, h)$.

$$
\begin{aligned}
\Pi(E_6, 1, 1) &= [E_6, 1];\\
\Pi(E_6(a_1), 1, 1) &= [E_6(a_1), 1];\\
\Pi(E_6(a_3), 1, 1) &= [E_6(a_3), +] + [E_6(a_3), -];\\
\Pi(E_6(a_3), 1, g_2) &= [E_6(a_3), +] - [E_6(a_3), -];\\
\Pi(E_6(a_3), g_2, 1) &= [A_1A_5, +] + [A_1A_5, -];\\
\Pi(E_6(a_3), g_2, g_2) &= [A_1A_5, +] - [A_1A_5, -];\\
\Pi(D_4, s_3, w_3) &= [D_4(71), 1] + \theta[D_4(71), \theta] + \theta^2[D_4(71), \theta^2];\\
\Pi(D_4, s_3, w_3^{-1}) &= [D_4(71), 1] + \theta^2[D_4(71), \theta] + \theta[D_4(71), \theta^2];\\
\Pi(D_4(a_1), 1, g_3) &= [D_4(a_1), 1] + [D_4(a_1), \varepsilon] - [D_4(a_1), r];\\
\Pi(D_4(a_1), z, g_3) &= [D_4(53), \mathbf{1}] + \theta[D_4(53), \theta] + \theta^2[D_4(53), \theta^2];\\
\Pi(D_4(a_1), z^2, g_3) &= \Pi(D_4(a_1), z, g_3^{-1}) = [D_4(53), \mathbf{1}] + \theta^2[D_4(53), \theta] + \theta[D_4(53), \theta^2];\\
\Pi(D_4(a_1), g_3, z^k g_3^l) &= \sum_{1 \leq i,j \leq 3} \theta^{il+jk}[3A_2, \theta^i \boxtimes \theta^j];\\
\Pi(A_2, s_{3,3}, w_{3,3}) &= \Pi(A_2, s_{3,3}, w_{3,3}^2) = [D_4(3311), \mathbf{1}] + [D_4(3311), \epsilon];
\end{aligned}
$$

7 Parahoric restriction tables

We now display the parahoric restriction tables obtained by applying the methods described in Section 4 together with the geometric data from $F_4(\mathbb{C})$ and $E_6(\mathbb{C})_{\mathrm{ad}}$ discussed in Sections 5 and 6. For each maximal parahoric subgroup K of $F_4(F)$ or $E_6(F)_{\mathrm{sc}}$, we display two tables. The first table contains the parahoric restrictions of the relevant unipotent representations arising in the virtual representations $\Pi(u,s,h)$. The second table then contains the parahoric restrictions of $\Pi(u,s,h)$ and is easily obtained from the first by computing the corresponding linear combinations. The first columns the second tables are the characters in trivial families; then we write each family with the special character always coming first. We omit the tables for the maximal hyperspecial subgroup K_0 of $F_4(F)$ for the tables are lengthy and the conjecture was already verified by Ciubotaru in [Ciu20]. From these tables, together with Lusztig's nonabelian Fourier transform, Theorems 5.1 and 6.1 can be easily verified

For all tables, we use the notation described in 3.6 for the Langlands labels parametrizing the unipotent representations of $G = F_4(F)$, coinciding with Reeder's notation in [Ree00]. The unipotent representations of finite reductive groups of Lie type are labelled as follows. For groups of type A_n, they are parametrized by partitions λ of $n+1$. For groups of type B_n/C_n, the principal series representations are parametrized by double partitions $[\lambda,\mu]$ of n. To understand these tables, it is enough to know that groups of type B_3/C_3 have two non-principal series representations θ_1 and θ_2, while groups of type B_4/C_4 have five non-principal series representations $\theta_1,\dots,\theta_5$. They all lie in the same representation series, arising from the unipotent cuspidal representation of their Levi subgroup of type B_2/C_2 and of all them are part of a family of size 4 of unipotent representations.

Finally, we use the following useful notation shorthand. A finite group of Lie type of type A_1 has only two unipotent characters; the trivial and the Steinberg. Thus, whenever the type of reductive quotient is the product of two simple types, one of which is A_1, we give the multiplicities for each character χ of the other type as a pair $a,\overline{b}$, where a and $\overline{b}$ are the multiplicities of $\chi\otimes\mathbf{1}_{A_1}$ and $\chi\otimes\mathrm{St}_{A_1}$, respectively.

7.1 Parahoric restriction tables for F_4

Unipotent B_4-types

Table 5: Parahoric restriction of Iwahori-spherical $F_4(F)$-representations with respect to K_1

	$[21^2,\cdot]$	$[1^4,\cdot]$	$[1^3,1]$	$[2,11]$	$[11,2]$	$[11,11]$	$[1,21]$	$[1,1^3]$	$[\cdot,31]$	$[\cdot,22]$	$[\cdot,21^2]$	$[\cdot,1^4]$
$[F_4,1]$	0	0	0	0	0	0	0	0	0	0	0	1
$[F_4(a_1),+]$	0	0	0	0	0	0	0	1	0	0	0	1
$[F_4(a_1),-]$	0	1	0	0	0	0	0	0	0	0	0	1
$[B_4,+]$	0	0	0	0	0	0	0	0	0	0	1	0
$[F_4(a_2),+]$	0	0	0	0	0	1	0	1	0	0	1	1
$[F_4(a_2),-]$	0	0	0	0	0	0	0	0	0	1	0	0
$[C_3A_1,+]$	0	0	1	0	0	0	0	1	0	0	1	1
$[B_4(711),+]$	0	0	0	0	0	0	1	1	0	0	1	0
$[B_4(711),-]$	0	0	0	0	0	1	0	0	0	1	0	1
$[F_4(a_3),4]$	0	0	1	1	1	1	1	2	0	0	1	1
$[F_4(a_3),31]$	1	1	1	0	0	1	0	1	0	0	0	1
$[F_4(a_3),22]$	0	0	0	1	0	0	0	1	0	0	0	0
$[F_4(a_3),211]$	0	1	0	0	0	0	0	0	0	0	0	0
$[C_3(42)A_1,++]$	1	0	1	1	1	1	1	2	0	0	2	1
$[C_3(42)A_1,+-]$	0	1	1	0	0	1	0	0	0	0	0	1
$[B_4(531),1]$	0	0	0	1	1	0	1	1	1	0	2	0
$[B_4(531),\varepsilon']$	0	0	0	0	0	0	0	0	1	0	0	0
$[B_4(531),\varepsilon'']$	1	0	0	0	0	0	0	0	0	0	1	0
$[2A_2,1]$	0	0	1	0	1	1	1	1	0	0	1	1
$[A_1A_3,1]$	0	0	0	0	1	0	1	1	1	0	1	0

Table 6: Parahoric restriction of non-Iwahori-spherical $F_4(F)$-representations with respect to K_1

	$\theta_1 = \left(\begin{smallmatrix} 0 & 1 & 2 & 3 & 4 \\ & 1 & 2 & & \end{smallmatrix}\right)$	$\theta_2 = \left(\begin{smallmatrix} 0 & 1 & 2 & 3 \\ & 2 & & \end{smallmatrix}\right)$	$\theta_3 = \left(\begin{smallmatrix} 0 & 1 & 2 & 4 \\ & 1 & & \end{smallmatrix}\right)$	$\theta_4 = \left(\begin{smallmatrix} 0 & 2 & 3 \\ & - & \end{smallmatrix}\right)$	$\theta_5 = \left(\begin{smallmatrix} 0 & 1 & 4 \\ & - & \end{smallmatrix}\right)$
$[B_4,-]$	1	0	0	0	0
$[C_3A_1,-]$	0	1	0	0	0
$[C_3(42)A_1,-+]$	1	0	0	0	0
$[B_4(531),r]$	1	0	1	0	0
$[A_1A_3,-]$	0	0	1	0	0

Table 7: Parahoric restrictions of the virtual characters $\Pi(u,s,h)$ with respect to K_1

	$[11,2]$	$[1,21]$	$[\cdot,1^4]$	$[1,1^3]$	$[1^4,\cdot]$	$[\cdot,21^2]$	θ_1	$[11,11]$	$[1^3,1]$	$[\cdot,22]$	θ_2	$[2,11]$	$[21^2,\cdot]$	$[\cdot,31]$	θ_3
$\Pi(F_4,1,1)$	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
$\Pi(F_4(a_1),1,1)$	0	0	2	1	1	0	0	0	0	0	0	0	0	0	0
$\Pi(F_4(a_1),1,g_2)$	0	0	0	1	-1	0	0	0	0	0	0	0	0	0	0
$\Pi(F_4(a_1),g_2,1)$	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0
$\Pi(F_4(a_1),g_2,g_2)$	0	0	0	0	0	1	-1	0	0	0	0	0	0	0	0
$\Pi(F_4(a_2),1,1)$	0	0	1	1	0	1	0	1	0	1	0	0	0	0	0
$\Pi(F_4(a_2),1,g_2)$	0	0	1	1	0	1	0	1	0	-1	0	0	0	0	0
$\Pi(F_4(a_2),g_2,1)$	0	0	1	1	0	1	0	0	1	0	1	0	0	0	0
$\Pi(F_4(a_2),g_2,g_2)$	0	0	1	1	0	1	0	0	1	0	-1	0	0	0	0
$\Pi(B_4(711),s,h)$	0	1	-1	1	0	1	0	-1	0	-1	0	0	0	0	0
$\Pi(F_4(a_3),1,1)$	1	1	4	7	6	1	0	4	4	0	0	3	3	0	0
$\Pi(F_4(a_3),1,g_2)$	1	1	2	3	0	1	0	2	2	0	0	1	1	0	0
$\Pi(F_4(a_3),1,g_2')$	1	1	0	3	-2	1	0	0	0	0	0	3	-1	0	0
$\Pi(F_4(a_3),1,g_3)$	1	1	1	1	0	1	0	1	1	0	0	0	0	0	0
$\Pi(F_4(a_3),1,g_4)$	1	1	0	1	0	1	0	0	0	0	0	1	-1	0	0
$\Pi(F_4(a_3),g_2,1)$	1	1	2	2	1	2	1	2	2	0	0	1	1	0	0
$\Pi(F_4(a_3),g_2,g_2)$	1	1	2	2	1	2	-1	2	2	0	0	1	1	0	0
$\Pi(F_4(a_3),g_2,\tau)$	1	1	0	2	-1	2	1	0	0	0	0	1	1	0	0
$\Pi(F_4(a_3),g_2,g_2')$	1	1	0	2	-1	2	-1	0	0	0	0	1	1	0	0
$\Pi(F_4(a_3),g_2',1)$	1	1	0	1	0	3	2	0	0	0	0	1	1	2	2
$\Pi(F_4(a_3),g_2',g_2)$	1	1	0	1	0	3	0	0	0	0	0	1	1	0	0
$\Pi(F_4(a_3),g_2',g_2')$	1	1	0	1	0	3	-2	0	0	0	0	1	1	2	-2
$\Pi(F_4(a_3),g_2',\gamma)$	1	1	0	1	0	1	0	0	0	0	0	1	-1	2	0
$\Pi(F_4(a_3),g_2',g_4)$	1	1	0	1	0	1	0	0	0	0	0	1	-1	0	0
$\Pi(F_4(a_3),g_3,h)$	1	1	1	1	0	1	0	1	1	0	0	0	0	0	0
$\Pi(F_4(a_3),g_4,h), h\in\{1,g_2'\}$	1	1	0	1	0	1	0	0	0	0	0	0	0	1	1
$\Pi(F_4(a_3),g_4,g_4^{\pm 1})$	1	1	0	1	0	1	0	0	0	0	0	0	0	1	-1

Table 8: Parahoric restriction of $F_4(F)$-representations with respect to K_2

	$[1^4]$	$[211]$	$[22]$	$[31]$	$[4]$
$[F_4, 1]$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_1), +]$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_1), -]$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[B_4, +]$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_2), +]$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_2), -]$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[C_3A_1, +]$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[B_4(711), +]$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$
$[B_4(711), -]$	$1, \bar{1}$	$0, \bar{1}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_3), 4]$	$2, \bar{3}$	$3, \bar{5}$	$1, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$
$[F_4(a_3), 31]$	$3, \bar{2}$	$2, \bar{2}$	$1, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_3), 22]$	$1, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$
$[F_4(a_3), 211]$	$1, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[C_3(42)A_1, ++]$	$2, \bar{3}$	$4, \bar{6}$	$1, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$
$[C_3(42)A_1, +-]$	$2, \bar{2}$	$1, \bar{1}$	$1, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[B_4(531), 1]$	$0, \bar{1}$	$2, \bar{4}$	$0, \bar{1}$	$1, \bar{3}$	$0, \bar{0}$
$[B_4(531), \varepsilon'']$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[B_4(531), \varepsilon']$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$
$[2A_2, 1]$	$1, \bar{3}$	$2, \bar{4}$	$1, \bar{1}$	$1, \bar{1}$	$0, \bar{0}$
$[A_1A_3, 1]$	$0, \bar{1}$	$1, \bar{3}$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$

Table 9: Parahoric restrictions of the virtual characters $\Pi(u, s, h)$ with respect to K_2

	$[1^4]$	$[211]$	$[22]$	$[31]$	$[4]$
$\Pi(F_4, 1, 1)$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(F_4(a_1), 1, 1)$	$2, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(F_4(a_1), 1, g_2)$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(F_4(a_1), g_2, 1)$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(F_4(a_1), g_2, g_2)$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(F_4(a_2), 1, 1)$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{2}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(F_4(a_2), 1, g_2)$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(F_4(a_2), g_2, 1)$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(F_4(a_2), g_2, g_2)$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(B_4(711), s, h)$	$-1, \bar{0}$	$1, \bar{1}$	$-1, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$
$\Pi(F_4(a_3), 1, 1)$	$16, \bar{9}$	$11, \bar{13}$	$4, \bar{1}$	$1, \bar{4}$	$0, \bar{0}$
$\Pi(F_4(a_3), 1, g_2)$	$4, \bar{5}$	$5, \bar{7}$	$2, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$
$\Pi(F_4(a_3), 1, g_2')$	$0, \bar{1}$	$3, \bar{5}$	$0, \bar{1}$	$1, \bar{4}$	$0, \bar{0}$
$\Pi(F_4(a_3), 1, g_3)$	$1, \bar{3}$	$2, \bar{4}$	$1, \bar{1}$	$1, \bar{1}$	$0, \bar{0}$
$\Pi(F_4(a_3), 1, g_4)$	$0, \bar{1}$	$1, \bar{3}$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_2, 1)$	$4, \bar{5}$	$5, \bar{7}$	$2, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_2, g_2)$	$4, \bar{5}$	$5, \bar{7}$	$2, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_2, \tau)$	$0, \bar{1}$	$3, \bar{5}$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_2, g_2')$	$0, \bar{1}$	$3, \bar{5}$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_2', 1)$	$0, \bar{1}$	$3, \bar{5}$	$0, \bar{1}$	$1, \bar{4}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_2', g_2)$	$0, \bar{1}$	$3, \bar{5}$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_2', \gamma)$	$0, \bar{1}$	$1, \bar{3}$	$0, \bar{1}$	$1, \bar{4}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_2', g_2')$	$0, \bar{1}$	$3, \bar{5}$	$0, \bar{1}$	$1, \bar{4}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_2', g_4)$	$0, \bar{1}$	$1, \bar{3}$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_3, h)$	$1, \bar{3}$	$2, \bar{4}$	$1, \bar{1}$	$1, \bar{1}$	$0, \bar{0}$
$\Pi(F_4(a_3), g_4, h)$	$0, \bar{1}$	$1, \bar{3}$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$

Unipotent $A_2 \times A_2$-types

Table 10: Parahoric restriction of $F_4(F)$-representations with respect to K_3

	$[1^3,1^3]$	$[1^3,21]$	$[1^3,3]$	$[21,1^3]$	$[21,21]$	$[21,3]$	$[3,1^3]$	$[3,21]$	$[3,3]$
$[F_4,1]$	1	0	0	0	0	0	0	0	0
$[F_4(a_1),+]$	1	1	0	1	0	0	0	0	0
$[F_4(a_1),-]$	0	1	0	0	0	0	0	0	0
$[B_4,+]$	1	0	0	1	0	0	0	0	0
$[F_4(a_2),+]$	2	2	0	2	1	0	0	0	0
$[F_4(a_2),-]$	0	0	0	1	0	0	0	0	0
$[C_3A_1,+]$	2	2	0	1	1	0	0	0	0
$[B_4(711),+]$	2	3	1	1	1	0	0	0	0
$[B_4(711),-]$	1	1	0	1	1	0	0	0	0
$[F_4(a_3),4]$	4	4	1	4	4	1	1	1	0
$[F_4(a_3),31]$	1	3	2	0	2	1	0	0	0
$[F_4(a_3),22]$	0	1	1	1	1	0	1	0	0
$[F_4(a_3),211]$	0	0	1	0	0	0	0	0	0
$[C_3(42)A_1,++]$	4	5	1	4	5	1	1	1	0
$[C_3(42)A_1,+-]$	1	2	1	0	1	1	0	0	0
$[B_4(531),1]$	3	2	0	5	3	0	2	1	0
$[B_4(531),\varepsilon'']$	0	1	0	0	1	0	0	0	0
$[B_4(531),\varepsilon']$	0	0	0	1	0	0	1	0	0
$[2A_2,1]$	4	3	0	3	3	1	0	1	0
$[A_1A_3,1]$	3	1	0	4	2	0	1	1	0

Table 11: Parahoric restrictions of the virtual characters $\Pi(u,s,h)$ with respect to K_3

	$[1^3,1^3]$	$[1^3,21]$	$[1^3,3]$	$[21,1^3]$	$[21,21]$	$[21,3]$	$[3,1^3]$	$[3,21]$	$[3,3]$
$\Pi(F_4,1,1)$	1	0	0	0	0	0	0	0	0
$\Pi(F_4(a_1),1,1)$	1	2	0	1	0	0	0	0	0
$\Pi(F_4(a_1),1,g_2)$	1	0	0	1	0	0	0	0	0
$\Pi(F_4(a_1),g_2,1)$	1	0	0	1	0	0	0	0	0
$\Pi(F_4(a_1),g_2,g_2)$	1	0	0	1	0	0	0	0	0
$\Pi(F_4(a_2),1,1)$	2	2	0	3	1	0	0	0	0
$\Pi(F_4(a_2),1,g_2)$	2	2	0	1	1	0	0	0	0
$\Pi(F_4(a_2),g_2,1)$	2	2	0	1	1	0	0	0	0
$\Pi(F_4(a_2),g_2,g_2)$	2	2	0	1	1	0	0	0	0
$\Pi(B_4(711),s,h)$	1	2	1	0	0	0	0	0	0
$\Pi(F_4(a_3),1,1)$	7	15	12	6	12	4	3	1	0
$\Pi(F_4(a_3),1,g_2)$	5	7	2	4	6	2	1	1	0
$\Pi(F_4(a_3),1,g_2')$	3	3	0	6	4	0	3	1	0
$\Pi(F_4(a_3),1,g_3)$	4	3	0	3	3	1	0	1	0
$\Pi(F_4(a_3),1,g_4)$	3	1	0	4	2	0	1	1	0
$\Pi(F_4(a_3),g_2,1)$	5	7	2	4	6	2	1	1	0
$\Pi(F_4(a_3),g_2,g_2)$	5	7	2	4	6	2	1	1	0
$\Pi(F_4(a_3),g_2,\tau)$	3	3	0	4	4	0	1	1	0
$\Pi(F_4(a_3),g_2,g_2')$	3	3	0	4	4	0	1	1	0
$\Pi(F_4(a_3),g_2',1)$	3	3	0	6	4	0	3	1	0
$\Pi(F_4(a_3),g_2',g_2)$	3	3	0	4	4	0	1	1	0
$\Pi(F_4(a_3),g_2',\gamma)$	3	1	0	6	2	0	3	1	0
$\Pi(F_4(a_3),g_2',g_2')$	3	3	0	6	4	0	3	1	0
$\Pi(F_4(a_3),g_2',g_4)$	3	1	0	4	2	0	1	1	0
$\Pi(F_4(a_3),g_3,h)$	4	3	0	3	3	1	0	1	0
$\Pi(F_4(a_3),g_4,h)$	3	1	0	4	2	0	1	1	0

Iwahori-spherical $A_1 \times C_3$-types

Table 12: Parahoric restriction of Iwahori-spherical $F_4(F)$-representations with respect to K_4

	$[\cdot, 1^3]$	$[1^3, \cdot]$	$[11, 1]$	$[1, 11]$	$[21, \cdot]$	$[\cdot, 21]$	$[2, 1]$	$[1, 2]$	$[\cdot, 3]$	$[3, \cdot]$
$[F_4, 1]$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_1), +]$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_1), -]$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[B_4, +]$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_2), +]$	$1, \bar{2}$	$0, \bar{0}$	$0, \bar{1}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_2), -]$	$0, \bar{1}$	$1, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[C_3A_1, +]$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[B_4(711), +]$	$2, \bar{2}$	$1, \bar{1}$	$0, \bar{1}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[B_4(711), -]$	$0, \bar{1}$	$0, \bar{0}$	$1, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_3), 4]$	$2, \bar{2}$	$0, \bar{1}$	$1, \bar{2}$	$2, \bar{3}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{1}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_3), 31]$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$1, \bar{2}$	$0, \bar{1}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$
$[F_4(a_3), 22]$	$1, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[F_4(a_3), 211]$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$
$[C_3(42)A_1, ++]$	$2, \bar{2}$	$0, \bar{1}$	$1, \bar{2}$	$2, \bar{3}$	$0, \bar{1}$	$2, \bar{2}$	$0, \bar{1}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[C_3(42)A_1, +-]$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$
$[B_4(531), 1]$	$3, \bar{2}$	$1, \bar{2}$	$1, \bar{2}$	$2, \bar{2}$	$0, \bar{1}$	$1, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[B_4(531), \varepsilon'']$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[B_4(531), \varepsilon']$	$1, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[2A_2, 1]$	$1, \bar{2}$	$0, \bar{1}$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[A_1A_3, 1]$	$2, \bar{2}$	$1, \bar{2}$	$1, \bar{2}$	$1, \bar{1}$	$0, \bar{0}$	$1, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$

Table 13: Parahoric restriction of non-Iwahori-spherical $F_4(F)$-representations with respect to K_4

	$\overline{\theta_1} = \left(\begin{smallmatrix} 0 & 1 & 2 & 3 \\ & 1 & & \end{smallmatrix}\right) \otimes \mathrm{St}$	$\theta_1 = \left(\begin{smallmatrix} 0 & 1 & 2 & 3 \\ & 1 & & \end{smallmatrix}\right) \otimes \mathbf{1}$	$\overline{\theta_2} = \left(\begin{smallmatrix} 0 & 1 & 3 \\ & - & \end{smallmatrix}\right) \otimes \mathrm{St}$	$\theta_2 = \left(\begin{smallmatrix} 0 & 1 & 3 \\ & - & \end{smallmatrix}\right) \otimes \mathbf{1}$
$[B_4, -]$	1	0	0	0
$[C_3A_1, -]$	0	1	0	0
$[C_3(42)A_1, -+]$	0	0	1	0
$[B_4(531), r]$	1	1	1	0
$[A_1A_3, -]$	1	1	0	0

Table 14: Parahoric restrictions of the virtual characters $\Pi(u,s,h)$ with respect to K_4

	$[1,2]$	$[11,1]$	$[\cdot,1^3]$	$[1,11]$	$[1^3,\cdot]$	$[\cdot,21]$	θ_1	$[2,1]$	$[21,\cdot]$	$[\cdot,3]$	θ_2
$\Pi(F_4,1,1)$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_1),1,1)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_1),1,g_2)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{1}$	$0,\bar{0}$	$0,-\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_1),g_2,1)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_1),g_2,g_2)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,-\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_2),1,1)$	$0,\bar{0}$	$0,\bar{1}$	$1,\bar{3}$	$1,\bar{1}$	$1,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_2),1,g_2)$	$0,\bar{0}$	$0,\bar{1}$	$1,\bar{1}$	$1,\bar{1}$	$-1,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_2),g_2,1)$	$0,\bar{0}$	$0,\bar{1}$	$1,\bar{1}$	$0,\bar{1}$	$0,\bar{0}$	$1,\bar{1}$	$1,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_2),g_2,g_2)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{1}$	$0,\bar{0}$	$1,\bar{1}$	$-1,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(B_4(711),s,h)$	$0,\bar{0}$	$-1,\bar{1}$	$2,\bar{1}$	$1,\bar{0}$	$1,\bar{1}$	$0,-\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_3),1,1)$	$4,\bar{4}$	$1,\bar{2}$	$4,\bar{2}$	$4,\bar{8}$	$0,\bar{1}$	$4,\bar{7}$	$0,\bar{0}$	$0,\bar{6}$	$0,\bar{0}$	$0,\bar{6}$	$0,\bar{0}$
$\Pi(F_4(a_3),1,g_2)$	$2,\bar{2}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{4}$	$0,\bar{1}$	$2,\bar{3}$	$0,\bar{0}$	$0,\bar{2}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_3),1,g_2')$	$0,\bar{0}$	$1,\bar{2}$	$4,\bar{2}$	$4,\bar{4}$	$0,\bar{1}$	$0,-\bar{1}$	$0,\bar{0}$	$0,\bar{2}$	$0,\bar{0}$	$0,-\bar{2}$	$0,\bar{0}$
$\Pi(F_4(a_3),1,g_3)$	$1,\bar{1}$	$1,\bar{2}$	$1,\bar{2}$	$1,\bar{2}$	$0,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_3),1,g_4)$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$0,-\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_3),g_2,1)$	$2,\bar{2}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{4}$	$0,\bar{1}$	$2,\bar{3}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{1}$	$0,\bar{1}$	$0,\bar{1}$
$\Pi(F_4(a_3),g_2,g_2)$	$2,\bar{2}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{4}$	$0,\bar{1}$	$2,\bar{3}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{1}$	$0,\bar{1}$	$0,-\bar{1}$
$\Pi(F_4(a_3),g_2,\tau)$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$2,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{1}$	$0,-\bar{1}$	$0,\bar{1}$
$\Pi(F_4(a_3),g_2,g_2')$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$2,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{1}$	$0,-\bar{1}$	$0,-\bar{1}$
$\Pi(F_4(a_3),g_2',1)$	$0,\bar{0}$	$1,\bar{2}$	$4,\bar{2}$	$2,\bar{2}$	$2,\bar{3}$	$2,\bar{1}$	$2,\bar{2}$	$0,\bar{0}$	$0,\bar{2}$	$0,\bar{0}$	$0,\bar{2}$
$\Pi(F_4(a_3),g_2',g_2)$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$2,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{2}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_3),g_2',\gamma)$	$0,\bar{0}$	$1,\bar{2}$	$4,\bar{2}$	$2,\bar{2}$	$2,\bar{3}$	$0,-\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_3),g_2',g_2')$	$0,\bar{0}$	$1,\bar{2}$	$4,\bar{2}$	$2,\bar{2}$	$2,\bar{3}$	$2,\bar{1}$	$-2,-\bar{2}$	$0,\bar{0}$	$0,\bar{2}$	$0,\bar{0}$	$0,-\bar{2}$
$\Pi(F_4(a_3),g_2',g_4)$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$0,-\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_3),g_3,h)$	$1,\bar{1}$	$1,\bar{2}$	$1,\bar{2}$	$1,\bar{2}$	$0,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_3),g_4,1/g_2')$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$1,\bar{1}$	$1,\bar{2}$	$1,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(F_4(a_3),g_4,g_4^{\pm 1})$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$1,\bar{1}$	$1,\bar{2}$	$1,\bar{0}$	$-1,-\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$

7.2 Parahoric restriction tables for E_6

Unipotent E_6-types

Table 15: Parahoric restriction of unipotent $E_6(F)_{\mathrm{sc}}$ with respect to K_0, K_1 and K_6, where the triality action of Ω can be easily observed

	$J_0 \longrightarrow E_6$	$J_1 \longrightarrow E_6$	$J_6 \longrightarrow E_6$
$[E_6, 1]$	$\phi_{1,36}$		
$[E_6(a_1), 1]$	$\phi_{1,36} + \phi_{6,25}$		
$[E_6(a_3), +]$	$\phi_{1,36} + \phi_{6,25} + \phi_{20,20} + \phi_{30,15}$		
$[E_6(a_3),]$	$\phi_{6,25} + \phi_{15,17}$		
$[A_1A_5, +]$	$\phi_{1,36} + \phi_{15,16} + \phi_{20,20}$		
$[A_1A_5, -]$	$D_4[\epsilon]$		
$[D_4(71), 1]$	$\phi_{1,36} + \phi_{15,17} + \phi_{20,20} + \phi_{24,12} + \phi_{30,15}$	$\phi_{6,25} + \phi_{20,20} + \phi_{64,13}$	$\phi_{6,25} + \phi_{20,20} + \phi_{64,13}$
$[D_4(71), \theta]$	$\phi_{6,25} + \phi_{20,20} + \phi_{64,13}$	$\phi_{6,25} + \phi_{20,20} + \phi_{64,13}$	$\phi_{1,36} + \phi_{15,17} + \phi_{20,20} + \phi_{24,12} + \phi_{30,15}$
$[D_4(71), \theta^2]$	$\phi_{6,25} + \phi_{20,20} + \phi_{64,13}$	$\phi_{1,36} + \phi_{15,17} + \phi_{20,20} + \phi_{24,12} + \phi_{30,15}$	$\phi_{6,25} + \phi_{20,20} + \phi_{64,13}$
$[D_4(a_1), 1]$	$\phi_{1,36} + \phi_{6,25} + \phi_{15,16} + 3\phi_{20,20} + \phi_{24,12} + 2\phi_{30,15} + \phi_{80,7} + 2\phi_{60,11} + 2\phi_{64,13} + \phi_{81,10}$		
$[D_4(a_1), r]$	$\phi_{6,25} + \phi_{15,17} + \phi_{20,20} + \phi_{30,15} + \phi_{90,8} + 2\phi_{64,13} + \phi_{81,10}$		
$[D_4(a_1), \epsilon]$	$\phi_{20,10} + \phi_{15,17}$		
$[D_4(53), 1]$	$\phi_{1,36} + \phi_{6,25} + \phi_{20,10} + \phi_{15,17} + \phi_{15,16} + \phi_{20,20} + \phi_{24,12} + 2\phi_{30,15} + \phi_{80,7} + 2\phi_{64,13} + \phi_{81,10}$	short	short
$[D_4(53), \theta]$	$\phi_{6,25} + \phi_{15,17} + 2\phi_{20,20} + \phi_{30,15} + \phi_{90,8} + \phi_{60,11} + 2\phi_{64,13} + \phi_{81,10}$	short	long
$[D_4(53), \theta^2]$	$\phi_{6,25} + \phi_{15,17} + 2\phi_{20,20} + \phi_{30,15} + \phi_{90,8} + \phi_{60,11} + 2\phi_{64,13} + \phi_{81,10}$	long	short
$[3A_2, 1 \boxtimes 1]$	$\phi_{1,36} + \phi_{10,9} + \phi_{15,16} + \phi_{24,12} + \phi_{30,15}$	$\phi_{20,20} + \phi_{60,11}$	$\phi_{20,20} + \phi_{60,11}$
$[3A_2, \theta \boxtimes 1]$	$E_6[\theta]$	0	0
$[3A_2, \theta^2 \boxtimes 1]$	$E_6[\theta^2]$	0	0
$[3A_2, 1 \boxtimes \theta]$	$\phi_{20,20} + \phi_{60,11}$	$\phi_{1,36} + \phi_{10,9} + \phi_{15,16} + \phi_{24,12} + \phi_{30,15}$	$\phi_{20,20} + \phi_{60,11}$
$[3A_2, \theta \boxtimes \theta]$	0	$E_6[\theta]$	0
$[3A_2, \theta^2 \boxtimes \theta]$	0	$E_6[\theta^2]$	0
$[3A_2, 1 \boxtimes \theta^2]$	$\phi_{20,20} + \phi_{60,11}$	$\phi_{20,20} + \phi_{60,11}$	$\phi_{1,36} + \phi_{10,9} + \phi_{15,16} + \phi_{24,12} + \phi_{30,15}$
$[3A_2, \theta \boxtimes \theta^2]$	0	0	$E_6[\theta]$
$[3A_2, \theta^2 \boxtimes \theta^2]$	0	0	$E_6[\theta^2]$
$[D_4(3311), 1]$	$\phi_{1,36} + \phi_{10,9} + 3\phi_{6,25} + \phi_{20,10} + 3\phi_{15,17} + 4\phi_{15,16} + 8\phi_{20,20} + \phi_{24,6} + 5\phi_{24,12} + \phi_{30,3} + 8\phi_{30,15} + 6\phi_{60,8} + 9\phi_{80,7} + 9\phi_{90,8} + 3\phi_{60,5} + 11\phi_{60,11} + 2\phi_{64,4} + 14\phi_{64,13} + 6\phi_{81,6} + 12\phi_{81,10}$		
$[D_4(3311), \varepsilon]$	$\phi_{6,25} + 3\phi_{20,10} + \phi_{15,5} + 3\phi_{15,17} + \phi_{15,16} + 2\phi_{20,20} + \phi_{24,6} + \phi_{24,12} + 3\phi_{30,15} + 2\phi_{60,8} + 3\phi_{80,7} + 6\phi_{90,8} + \phi_{60,5} + 3\phi_{60,11} + 2\phi_{64,4} + 6\phi_{64,13} + 3\phi_{81,6} + 6\phi_{81,10}$		

Table 16: Parahoric restrictions of the virtual characters $\Pi(u,s,h)$ with respect to K_0

	$\phi_{80,7}$	$\phi_{90,8}$	$\phi_{20,10}$	$\phi_{60,8}$	$D_4[r]$	$\phi_{10,9}$	$E_6[\theta]$	$E_6[\theta^2]$	$\phi_{30,15}$	$\phi_{15,17}$	$\phi_{15,16}$	$D_4[\epsilon]$	$\phi_{30,3}$	$\phi_{15,5}$	$\phi_{15,4}$	$D_4[1]$	Fourier-stable
$\Pi(E_6,1,1)$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	$\phi_{1,36}$
$\Pi(E_6(a_1),1,1)$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	$\phi_{1,36}+\phi_{6,25}$
$\Pi(E_6(a_3),1,1)$	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	$\Phi_{E_6(a_3)}$
$\Pi(E_6(a_3),1,g_2)$	0	0	0	0	0	0	0	0	1	-1	0	0	0	0	0	0	$\phi_{1,36}+\phi_{20,20}$
$\Pi(E_6(a_3),g_2,1)$	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	$\phi_{1,36}+\phi_{20,20}$
$\Pi(E_6(a_3),g_2,g_2)$	0	0	0	0	0	0	0	0	0	0	1	-1	0	0	0	0	$\phi_{1,36}+\phi_{20,20}$
$\Pi(D_4,s_3,w_3)$	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	Φ_{D_4}
$\Pi(D_4,s_3,w_3^2)$	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	Φ_{D_4}
$\Pi(D_4(a_1),1,g_3)$	1	-1	1	0	0	0	0	0	1	1	0	0	0	0	0	0	$\Phi^1_{D_4(a_1)}$
$\Pi(D_4(a_1),z,g_3)$	1	-1	1	0	0	0	0	0	1	1	0	0	0	0	0	0	$\Phi^2_{D_4(a_1)}$
$\Pi(D_4(a_1),z^2,g_3)$	1	-1	1	0	0	0	0	0	1	1	0	0	0	0	0	0	$\Phi^3_{D_4(a_1)}$
$\Pi(D_4(a_1),g_3,1)$	0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	$\Phi^1_{D_4(a_1)}$
$\Pi(D_4(a_1),g_3,z)$	0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	$\Phi^2_{D_4(a_1)}$
$\Pi(D_4(a_1),g_3,z^2)$	0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	$\Phi^3_{D_4(a_1)}$
$\Pi(A_2,s_{3,3},w_{3,3})$	12	15	4	8	0	1	0	0	11	6	5	0	1	1	0	0	Φ_{A_2}

Where, in the above table, all Φ_- are Fourier-stable combinations of the irreducible characters of $E_6(F)$ that are easily calculated from Table 15.

Unipotent $A_1 \times A_5$-types

Table 17: Parahoric restriction of Iwahori-spherical $E_6(F)$-representations with respect to K_2

V	$[1^6]$	$[21^4]$	$[2^21^2]$	$[222]$	$[31^3]$	$[321]$	$[333]$	$[41^2]$	$[42]$	$[51]$	$[6]$
$[E_6, 1]$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[E_6(a_1), 1]$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[E_6(a_3), +]$	$2, \bar{2}$	$1, \bar{3}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[E_6(a_3), -]$	$1, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[A_1A_5, +]$	$1, \bar{2}$	$0, \bar{2}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(71), 1]$	$1, \bar{2}$	$2, \bar{2}$	$1, \bar{2}$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(71), \theta]$	$1, \bar{1}$	$2, \bar{3}$	$1, \bar{2}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(71), \theta^2]$	$1, \bar{1}$	$2, \bar{3}$	$1, \bar{2}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(a_1), 1]$	$3, \bar{5}$	$6, \bar{10}$	$7, \bar{10}$	$2, \bar{5}$	$4, \bar{6}$	$4, \bar{6}$	$1, \bar{1}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(a_1), \epsilon]$	$0, \bar{0}$	$1, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(a_1), r]$	$2, \bar{1}$	$5, \bar{5}$	$4, \bar{4}$	$1, \bar{1}$	$4, \bar{6}$	$2, \bar{4}$	$0, \bar{0}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(53), 1]$	$3, \bar{3}$	$5, \bar{6}$	$5, \bar{6}$	$2, \bar{3}$	$5, \bar{7}$	$2, \bar{4}$	$1, \bar{1}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(53), \theta]$	$2, \bar{2}$	$6, \bar{7}$	$5, \bar{6}$	$1, \bar{2}$	$4, \bar{6}$	$3, \bar{5}$	$0, \bar{0}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(53), \theta^2]$	$2, \bar{2}$	$6, \bar{7}$	$5, \bar{6}$	$1, \bar{2}$	$4, \bar{6}$	$3, \bar{5}$	$0, \bar{0}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[3A_2, 1 \boxtimes 1]$	$1, \bar{2}$	$0, \bar{1}$	$1, \bar{2}$	$1, \bar{2}$	$1, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[3A_2, 1 \boxtimes \theta]$	$0, \bar{1}$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[3A_2, 1 \boxtimes \theta^2]$	$0, \bar{1}$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(3311), 1]$	$1, \bar{0}$	$8, \bar{6}$	$24, \bar{18}$	$17, \bar{14}$	$31, \bar{27}$	$70, \bar{58}$	$29, \bar{23}$	$55, \bar{47}$	$60, \bar{51}$	$44, \bar{37}$	$13, \bar{11}$
$[D_4(3311), \epsilon]$	$0, \bar{0}$	$6, \bar{3}$	$12, \bar{9}$	$6, \bar{4}$	$21, \bar{15}$	$30, \bar{26}$	$9, \bar{10}$	$27, \bar{25}$	$21, \bar{21}$	$15, \bar{17}$	$3, \bar{4}$

Table 18: Parahoric restriction of the virtual characters $\Pi(u, s, h)$ with respect to K_2

	$[1^6]$	$[21^4]$	$[2^21^2]$	$[222]$	$[31^3]$	$[321]$	$[333]$	$[41^2]$	$[42]$	$[51]$	$[6]$
$\Pi(E_6, 1, 1)$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(E_6(a_1), 1, 1)$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(E_6(a_3), 1, 1)$	$3, \bar{2}$	$2, \bar{4}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{2}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(E_6(a_3), 1, g_2)$	$1, \bar{2}$	$0, \bar{2}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(E_6(a_3), g_2, 1)$	$1, \bar{2}$	$0, \bar{2}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(E_6(a_3), g_2, g_2)$	$1, \bar{2}$	$0, \bar{2}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(D_4, s_3, w_3)$	$0, \bar{1}$	$0, -\bar{1}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{1}$	$0, -\bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(D_4, s_3, w_3^{-1})$	$0, \bar{1}$	$0, -\bar{1}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{1}$	$0, -\bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(D_4(a_1), 1, g_3)$	$1, \bar{4}$	$2, \bar{5}$	$3, \bar{6}$	$1, \bar{4}$	$1, \bar{1}$	$2, \bar{2}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(D_4(a_1), z, g_3)$	$1, \bar{1}$	$-1, -\bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$1, \bar{1}$	$-1, -\bar{1}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(D_4(a_1), z, g_3^{-1})$	$1, \bar{1}$	$-1, -\bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$1, \bar{1}$	$-1, -\bar{1}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(D_4(a_1), g_3, 1)$	$1, \bar{4}$	$2, \bar{5}$	$3, \bar{6}$	$1, \bar{4}$	$1, \bar{1}$	$2, \bar{2}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(D_4(a_1), g_3, z)$	$1, \bar{1}$	$-1, -\bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$1, \bar{1}$	$-1, -\bar{1}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(D_4(a_1), g_3, z^2)$	$1, \bar{1}$	$-1, -\bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$1, \bar{1}$	$-1, -\bar{1}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$\Pi(A_2, s_{3,3}, w_{3,3})$	$1, \bar{0}$	$14, \bar{9}$	$36, \bar{27}$	$23, \bar{18}$	$52, \bar{42}$	$100, \bar{84}$	$38, \bar{33}$	$82, \bar{72}$	$81, \bar{72}$	$59, \bar{54}$	$16, \bar{15}$

Table 19: Parahoric restriction of Iwahori spherical $E_6(F)$-representations with respect to K_3

V	$[1^6]$	$[21^4]$	$[2^21^2]$	$[222]$	$[31^3]$	$[321]$	$[333]$	$[41^2]$	$[42]$	$[51]$	$[6]$
$[E_6, 1]$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[E_6(a_1), 1]$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[E_6(a_3), +]$	$2, \bar{2}$	$1, \bar{3}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[E_6(a_3), -]$	$1, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[A_1A_5, +]$	$1, \bar{2}$	$0, \bar{2}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(71), 1]$	$1, \bar{1}$	$2, \bar{3}$	$1, \bar{2}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(71), \theta]$	$1, \bar{1}$	$2, \bar{3}$	$1, \bar{2}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(71), \theta^2]$	$1, \bar{2}$	$2, \bar{2}$	$1, \bar{2}$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(a_1), 1]$	$3, \bar{5}$	$6, \bar{10}$	$7, \bar{10}$	$2, \bar{5}$	$4, \bar{6}$	$4, \bar{6}$	$1, \bar{1}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(a_1), \epsilon]$	$0, \bar{0}$	$1, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(a_1), r]$	$2, \bar{1}$	$5, \bar{5}$	$4, \bar{4}$	$1, \bar{1}$	$4, \bar{6}$	$2, \bar{4}$	$0, \bar{0}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(53), 1]$	$2, \bar{2}$	$6, \bar{7}$	$5, \bar{6}$	$1, \bar{2}$	$4, \bar{6}$	$3, \bar{5}$	$0, \bar{0}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(53), \theta]$	$2, \bar{2}$	$6, \bar{7}$	$5, \bar{6}$	$1, \bar{2}$	$4, \bar{6}$	$3, \bar{5}$	$0, \bar{0}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(53), \theta^2]$	$3, \bar{3}$	$5, \bar{6}$	$5, \bar{6}$	$2, \bar{3}$	$5, \bar{7}$	$2, \bar{4}$	$1, \bar{1}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[3A_2, 1 \boxtimes 1]$	$0, \bar{1}$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[3A_2, 1 \boxtimes \theta]$	$1, \bar{2}$	$0, \bar{1}$	$1, \bar{2}$	$1, \bar{2}$	$1, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[3A_2, 1 \boxtimes \theta^2]$	$0, \bar{1}$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(3311), 1]$	$1, \bar{0}$	$8, \bar{6}$	$24, \bar{18}$	$17, \bar{14}$	$31, \bar{27}$	$70, \bar{58}$	$29, \bar{23}$	$55, \bar{47}$	$60, \bar{51}$	$44, \bar{37}$	$13, \bar{11}$
$[D_4(3311), \epsilon]$	$0, \bar{0}$	$6, \bar{3}$	$12, \bar{9}$	$6, \bar{4}$	$21, \bar{15}$	$30, \bar{26}$	$9, \bar{10}$	$27, \bar{25}$	$21, \bar{21}$	$15, \bar{17}$	$3, \bar{4}$

Table 20: Parahoric restriction of Iwahori-spherical $E_6(F)$-representations with respect to K_5

V	$[1^6]$	$[21^4]$	$[2^21^2]$	$[222]$	$[31^3]$	$[321]$	$[333]$	$[41^2]$	$[42]$	$[51]$	$[6]$
$[E_6, 1]$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[E_6(a_1), 1]$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[E_6(a_3), +]$	$2, \bar{2}$	$1, \bar{3}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[E_6(a_3), -]$	$1, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[A_1A_5, +]$	$1, \bar{2}$	$0, \bar{2}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(71), 1]$	$1, \bar{1}$	$2, \bar{3}$	$1, \bar{2}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(71), \theta]$	$1, \bar{2}$	$2, \bar{2}$	$1, \bar{2}$	$0, \bar{1}$	$1, \bar{2}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(71), \theta^2]$	$1, \bar{1}$	$2, \bar{3}$	$1, \bar{2}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(a_1), 1]$	$3, \bar{5}$	$6, \bar{10}$	$7, \bar{10}$	$2, \bar{5}$	$4, \bar{6}$	$4, \bar{6}$	$1, \bar{1}$	$1, \bar{1}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(a_1), \epsilon]$	$0, \bar{0}$	$1, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(a_1), r]$	$2, \bar{1}$	$5, \bar{5}$	$4, \bar{4}$	$1, \bar{1}$	$4, \bar{6}$	$2, \bar{4}$	$0, \bar{0}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(53), 1]$	$2, \bar{2}$	$6, \bar{7}$	$5, \bar{6}$	$1, \bar{2}$	$4, \bar{6}$	$3, \bar{5}$	$0, \bar{0}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(53), \theta]$	$3, \bar{3}$	$5, \bar{6}$	$5, \bar{6}$	$2, \bar{3}$	$5, \bar{7}$	$2, \bar{4}$	$1, \bar{1}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(53), \theta^2]$	$2, \bar{2}$	$6, \bar{7}$	$5, \bar{6}$	$1, \bar{2}$	$4, \bar{6}$	$3, \bar{5}$	$0, \bar{0}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$
$[3A_2, 1 \boxtimes 1]$	$0, \bar{1}$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[3A_2, 1 \boxtimes \theta]$	$0, \bar{1}$	$1, \bar{2}$	$1, \bar{2}$	$0, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[3A_2, 1 \boxtimes \theta^2]$	$1, \bar{2}$	$0, \bar{1}$	$1, \bar{2}$	$1, \bar{2}$	$1, \bar{1}$	$0, \bar{0}$	$1, \bar{1}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$	$0, \bar{0}$
$[D_4(3311), 1]$	$1, \bar{0}$	$8, \bar{6}$	$24, \bar{18}$	$17, \bar{14}$	$31, \bar{27}$	$70, \bar{58}$	$29, \bar{23}$	$55, \bar{47}$	$60, \bar{51}$	$44, \bar{37}$	$13, \bar{11}$
$[D_4(3311), \epsilon]$	$0, \bar{0}$	$6, \bar{3}$	$12, \bar{9}$	$6, \bar{4}$	$21, \bar{15}$	$30, \bar{26}$	$9, \bar{10}$	$27, \bar{25}$	$21, \bar{21}$	$15, \bar{17}$	$3, \bar{4}$

Unipotent $A_2 \times A_2 \times A_2$-types

Table 21: Parahoric restriction of Iwahori-spherical $E_6(F)$-representations with respect to K_4

V	$[\varepsilon\varepsilon\varepsilon]$	$[\varepsilon\varepsilon r]$	$[\varepsilon\varepsilon 1]$	$[\varepsilon rr]$	$[\varepsilon r1]$	$[\varepsilon 11]$	$[rrr]$	$[rr1]$	$[r11]$	$[111]$
$[E_6, 1]$	1	0	0	0	0	0	0	0	0	0
$[E_6(a_1), 1]$	1	1	0	0	0	0	0	0	0	0
$[E_6(a_3), +]$	4	3	1	2	0	0	1	0	0	0
$[E_6(a_3), -]$	0	1	1	1	0	0	0	0	0	0
$[A_1A_5, +]$	4	2	0	1	0	0	1	0	0	0
$[D_4(71), 1]$	3	3,4,3	1,0,1	3,4,3	1,0,1,0,1,1	0,1,0	2	0	0	0
$[D_4(71), \theta]$	3	4,3,3	0,1,1	3,3,4	0,1,0,1,1,1	0,0,1	2	0	0	0
$[D_4(71), \theta^2]$	3	3,3,4	1,1,0	4,3,3	1,1,1,1,0,0	1,0,0	2	0	0	0
$[D_4(a_1), 1]$	15	15	3	17	4	1	19	4	1	0
$[D_4(a_1), \epsilon]$	0	0	1	1	1	0	1	0	0	0
$[D_4(a_1), r]$	3	8	4	11	4	1	11	3	0	0
$[D_4(53), 1]$	7	10,11,10	4	13,14,13	5,4,4,4,4,5	1	14	3,4,3	0,1,0	0
$[D_4(53), \theta]$	7	10,10,11	4	14,13,13	4,4,5,5,4,4	1	14	3,3,4	1,0,0	0
$[D_4(53), \theta^2]$	7	11,10,10	4	13,13,14	4,5,4,4,5,4	1	14	4,3,3	0,0,1	0
$[3A_2, 1 \boxtimes 1]$	4	2,3,2	0	2,3,2	1,0,0,0,0,1	0	3	0,1,0	0,1,0	0
$[3A_2, 1 \boxtimes \theta]$	4	2,2,3	0	3,2,2	0,0,1,1,0,0	0	3	0,0,1	1,0,0	0
$[3A_2, 1 \boxtimes \theta^2]$	4	3,2,2	0	2,2,3	0,1,0,0,1,0	0	3	1,0,0	0,0,1	0
$[D_4(3311), 1]$	9	29	21	80	53	33	216	140	89	57
$[D_4(3311), \epsilon]$	3	13	11	40	29	19	96	60	33	15

Table 22: Parahoric restriction of the virtual characters $\Pi(u, s, h)$ with respect to K_4

Row Label	$[\varepsilon\varepsilon\varepsilon]$	$[\varepsilon\varepsilon r]$	$[\varepsilon\varepsilon 1]$	$[\varepsilon rr]$	$[\varepsilon r1]$	$[\varepsilon 11]$	$[rrr]$	$[rr1]$	$[r11]$	$[111]$
$\Pi(E_6, 1, 1)$	1	0	0	0	0	0	0	0	0	0
$\Pi(E_6(a_1), 1, 1)$	1	1	0	0	0	0	0	0	0	0
$\Pi(E_6(a_3), 1, 1)$	4	4	2	3	0	0	1	0	0	0
$\Pi(E_6(a_3), 1, g_2)$	4	2	0	1	0	0	1	0	0	0
$\Pi(E_6(a_3), g_2, 1)$	4	2	0	1	0	0	1	0	0	0
$\Pi(E_6(a_3), g_2, g_2)$	4	2	0	1	0	0	1	0	0	0
$\Pi(D_4, s_3, w_3)$	0	$\theta, 1, \theta^2$	$-\theta, -1, -\theta^2$	$\theta^2, 1, \theta$	$-\theta, -1, -\theta, -1, -\theta^2, -\theta^2$	$\theta^2, 1, \theta$	0	0	0	0
$\Pi(D_4, s_3, w_3^{-1})$	0	$\theta^2, 1, \theta$	$-\theta^2, -1, -\theta$	$\theta, 1, \theta^2$	$-\theta^2, -1, -\theta^2, -1, -\theta, -\theta$	$\theta, 1, \theta^2$	0	0	0	0
$\Pi(D_4(a_1), 1, g_3)$	12	7	0	7	1	0	9	1	1	0
$\Pi(D_4(a_1), z, g_3)$	0	$\theta^2, 1, \theta$	0	$\theta, 1, \theta^2$	$1, \theta^2, \theta, \theta, \theta^2, 1$	0	0	$\theta^2, 1, \theta$	$\theta, 1, \theta^2$	0
$\Pi(D_4(a_1), z, g_3^{-1})$	0	$\theta, 1, \theta^2$	0	$\theta^2, 1, \theta$	$1, \theta, \theta^2, \theta^2, \theta, 1$	0	0	$\theta, 1, \theta^2$	$\theta^2, 1, \theta$	0
$\Pi(D_4(a_1), g_3, 1)$	12	7	0	7	1	0	9	1	1	0
$\Pi(D_4(a_1), g_3, z)$	0	$\theta^2, 1, \theta$	0	$\theta, 1, \theta^2$	$1, \theta^2, \theta, \theta, \theta^2, 1$	0	0	$\theta^2, 1, \theta$	$\theta, 1, \theta^2$	0
$\Pi(D_4(a_1), g_3, z^2)$	0	$\theta, 1, \theta^2$	0	$\theta^2, 1, \theta$	$1, \theta, \theta^2, \theta^2, \theta, 1$	0	0	$\theta, 1, \theta^2$	$\theta^2, 1, \theta$	0
$\Pi(A_2, s_{3,3}, w_{3,3})$	12	42	32	120	82	52	312	200	122	72

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